

RESEARCH ARTICLE

Prototypical Few-Shot Learning for Histopathology Classification: Leveraging Foundation Models With Adapter Architectures

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ABSTRACT Histopathology is a critical tool for disease diagnosis and identifying cancer via microscopic tissue analysis. Traditional deep learning methods for histopathology often require extensive labeled data, which can be scarce and expensive. This study introduces a framework for the few-shot adaptation of self-supervised histopathology pretrained foundation models using multilayer perception adapters and convolutional adapters. An adapter comprising two linear or convolutional layers with nonlinear activation and residual connections transforms embeddings from foundation models for histopathology classification tasks. This study employs prototypical networks, SimpleShot, and bias-diminishing cosine similarity-based prototypical networks as few-shot learning algorithms. Comprehensive experiments are conducted across benchmark histopathology datasets: NCT, LC25000, Kather, and Camelyon17 Wilds. The results demonstrate that both adapter architectures consistently outperform the linear probe method, whereas multilayer perception adapters have an overall higher accuracy, especially when fine-tuned with five or more samples. The iBOT model with multilayer perception adapters fine-tuned using the bias-diminishing cosine similarity-based prototypical network algorithm achieved remarkable accuracy, reaching 95.04% on Camelyon17 Wilds and 96.55% on the NCT dataset with 20 images per class while using less than 0.002% of the dataset. These findings underscore the effectiveness of the proposed approach in addressing challenges posed by low-data regimes in the computer-aided histopathology domain and the potential for optimizing foundation models with minimal labeled data using prototypical few-shot algorithms.

INDEX TERMS

Adapter, bias-diminishing cosine similarity-based prototypical network (BDCSPN), breast cancer, Camelyon17, Camelyon17 wilds, colorectal cancer, convolutional (Conv) adapter, few-shot learning (FSL), foundation models, histopathology classification, Kather, LC25000, multilayer perception (MLP) adapter, NCT, PatchCamelyon (PCAM), prototypical network (ProtoNet), self-supervised learning (SSL), SimpleShot, multi-organ histopathology classification.

I. INTRODUCTION

Histopathology, the microscopic study of tissue, is crucial for diagnosing various diseases, such as cancer. This process involves analyzing histopathology images to identify subtle

morphological features indicative of disease, requiring expert knowledge and considerable time. However, this process is often subject to interobserver variability [1], [2]. Deep learning (DL) has demonstrated promise in automating and improving the efficiency of histopathology analysis [3], [4]. Despite this promise, training high-performing DL models requires a massive volume of labeled data [5], which is often

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scarce and expensive to obtain in the medical domain. This limitation underscores the need for innovative approaches that can employ existing knowledge and learn effectively from limited or unlabeled data. One promising solution is the foundation model [5] approach, involving pretraining models on large-scale unlabeled datasets and fine-tuning them for specific tasks using relatively small datasets [6].

Foundation models [5], [7], also known as large pretrained DL models, are trained on vast unlabeled datasets encompassing diverse data sources and modalities. The extensive unsupervised or self-supervised learning (SSL) allows DL models to learn general-purpose representations capturing intricate patterns and features [8], [9], [10], [11]. These representations can be transferred to downstream tasks, substantially improving performance and efficiency, even with limited labeled data [6]. In histopathological medical image analysis, foundation models have demonstrated great promise in identifying subtle morphological features indicative of disease via pretraining on large histopathology datasets, aiding pathologists in diagnosis and prognosis [7], [12]. Despite the success of foundation models, adapting them to rare, unseen histopathology tasks often requires additional labeled data, which can be time-consuming, expensive, and sometimes impossible to obtain. The crucial challenge is how to adapt foundation models with minimal labeled data efficiently. Few-shot learning (FSL) is a specialized branch of DL algorithms that addresses this challenge by enabling models to learn new concepts from a few labeled examples, mimicking the human ability to generalize from limited experience [13]. FSL methods allow models to learn new concepts from limited supervision, similar to how a child can recognize new birds without explicit instruction after seeing only a few of them.

The FSL encompasses a variety of strategies, including metric learning approaches, such as siamese networks [14], matching networks [15], and meta-learning algorithms (e.g., model-agnostic meta-learning [16]). Although recent studies have explored various FSL algorithms and the adaptability of the foundation models to solve diverse medical tasks, the potential of the prototypical network (ProtoNet) and many of its variants remains relatively underexplored, particularly in specific domains such as histopathology [17], [18]. This work proposes a novel and efficient adaptation mechanism to employ SSL-based foundation models for histopathology classifications. The focus is on efficiently adapting the learned feature embeddings from the foundation model [12], [19] to new histopathology classification tasks using limited labeled data. This work demonstrates the effectiveness of using small neural networks to adapt pretrained feature embeddings. The adapter models are designed to be lightweight and require minimal labeled data for fine-tuning. This paper makes the following contributions:

- **Novel adaptation framework:** We propose a novel framework, applying histopathology pretrained foundation models and adapting them to FSL tasks using an

efficient adapter module. This framework effectively employs pretrained feature representations with minimal labeled data.

- **Efficient learning scheme:** We employ a simple adapter module, modifying pretrained feature embeddings from foundation models to be specific to the target histopathology tasks. This adapter is fine-tuned using various prototypical FSL algorithms, allowing for quick adaptation with limited labeled examples and a deep dive into different prototypical learning-based FSL strategies.
- **Benchmark evaluation:** We conduct extensive experiments to validate the effectiveness of the proposed approach across histopathology classification tasks. The results demonstrate significant improvements in performance and generalization, even with limited labeled data, highlighting the potential of the proposed method in real-world medical applications.

This work aims to develop an efficient and practical framework for histopathology image analysis by combining SSL-based foundation models with FSL-based quick adaptation mechanisms, especially in limited labeled data scenarios.

II. RELATED WORK

The evolution of DL models in histopathology has followed two parallel tracks: (1) improving feature representation through pretraining on large datasets [7], [12], [17], and (2) developing efficient adaptation strategies for downstream tasks [20], [21], [22]. Although early studies have focused on the end-to-end training of convolutional neural networks [3], they require substantial labeled data, often unavailable in specialized medical domains. SSL has emerged as an effective approach to overcome this limitation, enabling neural networks to learn generalizable features from unlabeled histopathology data [23]. Recent advances in SSL-based foundation models [12], [17], [19] have emphasized large-scale pretraining on unlabeled data, drastically reducing dependence on extensive annotations. This paradigm facilitates knowledge transfer across diverse histopathology tasks, paving the way for more generalized, robust feature extractors. This section reviews the literature on various fine-tuning mechanisms for the quick adaptation of DL models for histopathology classification and establishes the background for the proposed method.

A. FINE-TUNE DEEP LEARNING MODELS

In the field of machine learning, a widely adopted and practical approach to fine-tuning involves employing a frozen feature extractor in conjunction with a trainable linear classification head [24]. The frozen feature extractor, typically a pretrained deep neural network, retains its learned weights to preserve the comprehensive image representations acquired during extensive pretraining. In addition to this feature extractor, a new, randomly initialized linear classifier is introduced. The classifier is fine-tuned to map the extracted

features to the target task classes. This method, commonly referred to as the linear probe approach, is efficient because it only updates the parameters of the classification head, making the process faster and less computationally demanding than fine-tuning the entire model. The reduced number of trainable parameters also mitigates the risk of overfitting on smaller target datasets.

This approach is widely applied in image classification tasks with pretrained DL models, such as residual networks (ResNet) [25] or vision transformers (ViT) [26], where a simple multilayer perceptron (MLP)-based classifier can yield impressive results on novel datasets. However, this approach may be limited in its ability to employ the power of the foundation models fully. The MLP classifier may not adequately capture subtle and complex high-dimensional feature representations when the model is applied to tasks that diverge from the original pretrained domain. This limitation suggests the need for more sophisticated adaptation mechanisms, improving the application of the pretrained model for diverse downstream tasks.

B. PARAMETER-EFFICIENT TRANSFER LEARNING AND ADAPTERS

Extensive research has focused on making transfer learning more efficient, especially in fine-tuning large pretrained models. Researchers have increasingly sought to reduce the number of trainable parameters to maintain the capabilities of the original models while easing the computational cost and risk of overfitting. Several parameter-efficient transfer learning [20] techniques have been proposed, such as prompt tuning [22], LoRA [21], etc. These methods aim to adapt pretrained models by modifying or updating a few parameters, often by adding low-rank matrices to the model weights [21]. These methods can mitigate the risk of catastrophic forgetting and offer effective solutions for multitask and continual learning using only a fraction of the parameters for each task. Recent work has demonstrated that fine-tuning selective layers of encoders on small datasets yields better results than full-model fine-tuning [27], [28].

Despite the advantages, these techniques require altering the weights or structure of the model when adapting to the target task, making it challenging to preserve the original model weights [20]. In contrast, employing a small network that dynamically interacts with the output features of the pretrained model could provide a more robust and efficient solution. The small network can be quickly fine-tuned to adapt and distill the relevant information from the pretrained features, whereas the original model weights remain unchanged [29]. Prior work has explored the use of such small networks to adapt pretrained multimodal models for downstream tasks [29], [30].

However, these adapter-based strategies typically address single-task or multimodal settings, and their synergy with FSL in histopathology has only recently begun to be investigated. For instance, Gao et al. [29] applied two

small networks to blend image and text information to improve natural image classification tasks. Lai et al. [30] employed a similar approach where a small fully connected module blends the text and image information and adapts the pretrained DL model to improve the performance on histopathology classification tasks. These small adapter networks differ from linear classifiers because they can capture complex feature interactions and adjust the model accordingly. This approach raises the following questions: What is the minimum number of image patches required to train this small network, and how can this be achieved efficiently? These questions motivate us to investigate the potential of FSL techniques for enabling efficient adaptation.

C. FEW-SHOT LEARNING FOR EFFICIENT ADAPTATION

The FSL method is a promising paradigm for efficiently adapting pretrained models to new tasks, where the aim is to learn new tasks from a few labeled examples [31]. In the context of histopathology, FSL could be beneficial because it can address the challenges of data scarcity and class imbalance often encountered in medical imaging tasks [32]. Prior work has explored FSL techniques for medical image analysis, demonstrating their potential to adapt pretrained models to novel tasks [13], [33], [34], [35], [36]. For instance, one study [13] found that FSL techniques in histopathology image analysis require significantly fewer training images to achieve comparable results to traditional fine-tuned neural networks. The study validated the transfer learning capabilities of FSL, demonstrating that the models could adapt from a more restricted source domain to a more generic target domain. Another study [33] developed a novel few-shot feature mapping network to solve various medical image segmentation tasks, such as optic-disc, cardiac structure, and polyp segmentation. In [35], researchers modified the FSL pipeline by integrating a strategy to select more meaningful images for breast and colon cancer classification tasks.

In [34], the researchers employed a prototypical FSL algorithm that applied a meta-learning approach, enabling the model to acquire new tasks with limited training data. Prototypical learning is a widely used FSL method, where a prototype represents each class, and new examples are classified based on their proximity to these prototypes [37]. Recent studies have explored similar FSL strategies for the adaptation of foundation models [17]. Researchers in [38] and [39] benchmarked the FSL methods under various domain shift scenarios to promote research on transferable DL models in medical imaging. Lee et al. [18] further extended this by benchmarking the ability of various foundation models to transfer medical domain knowledge with different FSL strategies. They also experimented with adaptation methods, including partial fine-tuning, full fine-tuning, and other strategies (e.g., linear probing and parameter-efficient transfer learning).

However, the existing literature lacks a focused and detailed analysis of prototypical FSL methods specifically

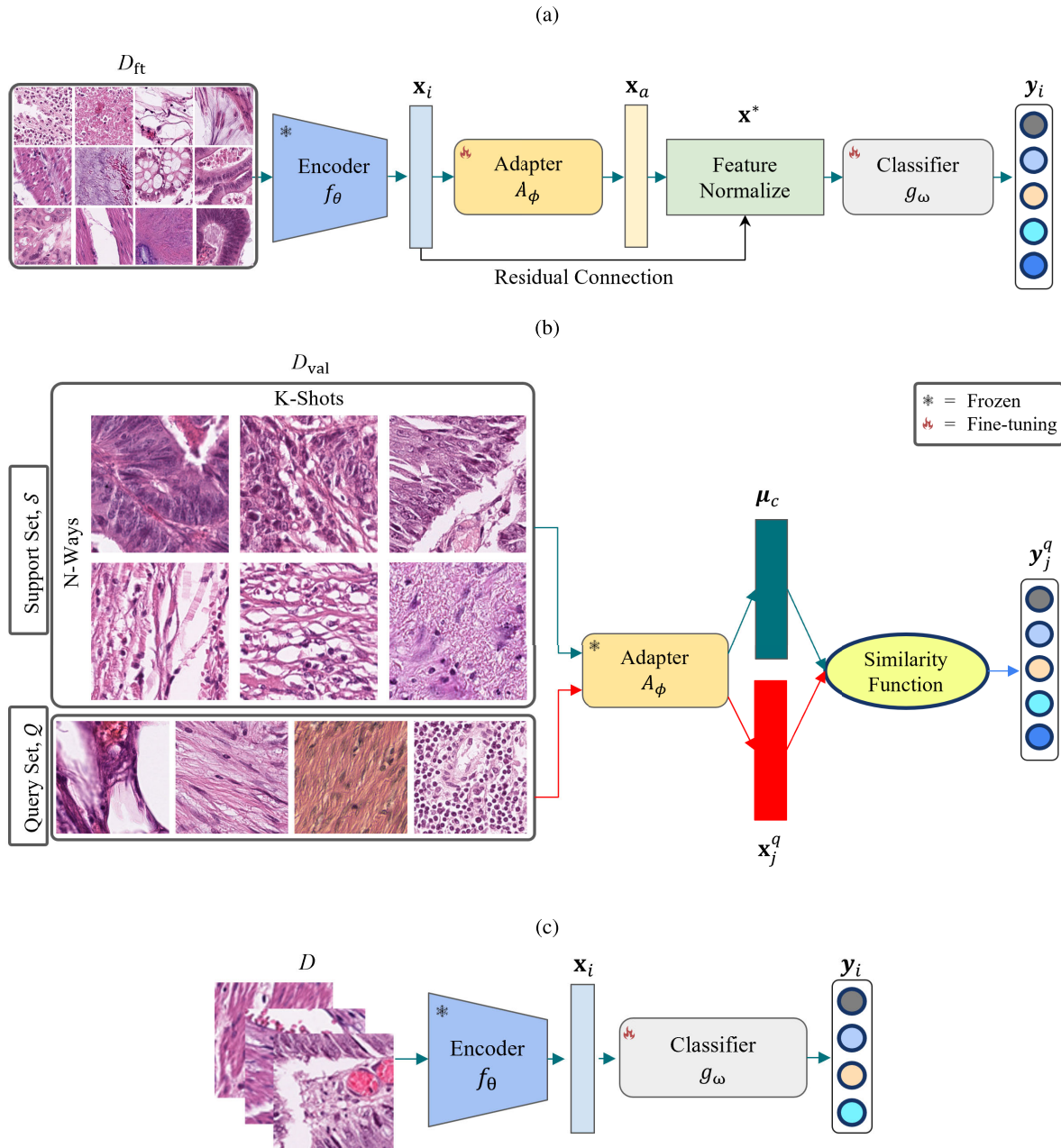


FIGURE 1. Histopathology classification framework. (a) Randomly sampled patches from D_{ft} are sequentially processed via the foundation model encoder $f_{\theta}(\cdot)$, learnable feature adapter $A_{\phi}(\cdot)$, and classification head $g_{\omega}(\cdot)$. Residual connections and feature normalization are applied before the classification head g_{ω} . (b) Prototype vectors are calculated using the support sets $\mathcal{S}$. Query samples $\mathcal{Q}$ are assigned classes based on their similarity to the mean prototypes using a learnable feature adapter $A_{\phi}(\cdot)$. (c) Linear probe experiment pipeline: Images are processed with a frozen feature extractor $f_{\theta}(\cdot)$ and classified using a fine-tuned $g_{\omega}(\cdot)$.

for adapting SSL-based foundation models to histopathology classification tasks in data-constrained scenarios. This work bridges this gap by validating prototypical FSL approaches from [37], [40], and [41] to adapt foundation models to histopathology classification.

III. METHODS

The downstream target dataset D , comprising image samples I , was partitioned into three subsets: D_{ft} , D_{val} , and D_{test} .

Fig. 1 presents the framework of the adapter-based method employed in this study. The image collection is denoted by $\mathbf{I} = \{I_1, I_2, \dots, I_i, \dots, I_D\}$, where I represents an individual patch image. In a deep neural network workflow, the images are processed using the network backbone/encoder $f_{\theta}(\cdot)$ (e.g., VGG [42], ResNet [25], transformer [43], etc.). The extracted feature vectors $\mathbf{x}$ are processed using the adapter network $A_{\phi}(\cdot)$ and classification head $g_{\omega}(\cdot)$ for classification tasks (see Fig. 1). Table 1 lists the symbols used throughout the paper.

TABLE 1. Notations.

Notation	Description	Notation	Description
I	Input histopathology image or patch	D	Dataset
$\mathbf{x}, \mathbf{y}$	Patch image in feature vector space, associated label	$D_{\text{ft}}, D_{\text{val}}, D_{\text{test}}$	Fine-tune, validation, and testing datasets, respectively
$\mathbf{x}_a$	Adapter-transformed feature vector	d	Dimension of feature space
$\mathbf{x}^*$	Normalized feature vector	$\mathcal{O}_{\text{ft}}$	Fine-tuned objective function
C	Number of classes	T	Number of patch images in fine-tuning
$\mathbb{R}$	Real number	$\boldsymbol{\mu}$	Mean prototype vector
$\mathbf{p}$	Probability vector over all classes	p	Predicted probability for a single class
$\mathcal{S}, \mathcal{Q}$	Support set, query set	M	Number of samples per class in $\mathcal{Q}$
N -way	Number of classes in few-shot validation tasks	K -shot	Number of samples per class in $\mathcal{S}$
H	Filter size in the Conv adapter	k	Kernel size in the Conv adapter
θ, ϕ, ω	Network parameters	τ	Temperature parameter
$f_{\theta}(\cdot)$	Frozen encoder or backbone model	$A_{\phi}(\cdot)$	Adapter network
$g_{\omega}(\cdot)$	Classification head	$\mathbf{W}_{\omega}$	Weight matrix of the classification head
L_{CE}	Cross-entropy loss	$\sigma(\cdot)$	Softmax activation function

In FSL, the D_{val} and D_{test} datasets are employed to formulate the labeled support set $\mathcal{S}$ and unlabeled query set $\mathcal{Q}$ for few-shot validation and testing. The sets $\mathcal{S}$ and $\mathcal{Q}$ share the same label space, analogous to the training and testing sets in standard classification. The few-shot characteristic stems from the limited number of labeled samples per class in the support set $\mathcal{S}$, presenting a challenge for training effective classification models.

In an N -way K -shot task setting, N novel classes are selected, with each class having K support images from the support set $\mathcal{S}$. The support set is defined as $\mathcal{S} = \{(\mathbf{x}_i^s, \mathbf{y}_i^s)\}_{i=1}^{N \times K}$, where $\mathbf{x}_i^s \in \mathbb{R}^d$ denotes the i th support image vector. Similarly, the query set $\mathcal{Q}$, containing M query images per class, is defined as $\mathcal{Q} = \{(\mathbf{x}_j^q, \mathbf{y}_j^q)\}_{j=1}^{N \times M}$. The total number of images in the support set is $N \times K$, and the total number of unlabeled query images is $N \times M$. The evaluation dataset D_{val} and testing dataset D_{test} are randomly sampled multiple times for FSL. The objective is to employ histopathology pretrained foundation models for efficient histopathology classification using FSL.

A. ADAPTER ARCHITECTURE

This study employs ResNet50 [25] and ViT-B [43] backbones as feature extractors, denoted by $f_{\theta}(\cdot)$. Following prior work, we employ two different adapter architectures: convolutional (Conv) and multilayer perceptrons (MLP) adapters [29], [30], [44]. Both adapter structures consist of a nonlinear activation (i.e., rectified linear units) and a residual connection that allows backbone features to bypass the adapter module. The weight matrices of the two network layers in the MLP adapter are represented as $\mathbf{W}_1^{\text{MLP}} \in \mathbb{R}^{d \times h}$ and $\mathbf{W}_2^{\text{MLP}} \in \mathbb{R}^{h \times d}$, where h represents the bottleneck layer dimension of the adapter network.

For the Conv adapter, we adopted the same approach as in [44], reshaping the d -dimensional feature vector into a square format, followed by pixel padding where necessary. The output from the ResNet50 feature encoder is reshaped into a 46×46 square feature map, whereas the ViT-B encoder produces a 28×28 map. After reshaping, the Conv adapter applied two convolutional layers represented by

$\mathbf{W}_1^{\text{Conv}} \in \mathbb{R}^{k \times k \times 1 \times H}$ and $\mathbf{W}_2^{\text{Conv}} \in \mathbb{R}^{k \times k \times H \times 1}$, where k denotes the kernel size, H represents the filter size, and 1 specifies the stride.

We reformatted the output from the adapter into a d -dimensional feature space before passing it to the classifier. In both cases, the adapter maps the output in the same dimension as the input, allowing the adapter output to be represented as $\mathbf{x}_a \in \mathbb{R}^d$.

B. ADAPTER FINE-TUNING MECHANISM

The pretrained encoder $f_{\theta}(\cdot)$ transforms each image sample I from D_{ft} into a high-dimensional feature vector $\mathbf{x} \in \mathbb{R}^d$, where d denotes the feature dimension. This transformation is expressed in Eq. (1):

$$\mathbf{x} = f_{\theta}(I). \quad (1)$$

During the fine-tuning stage, the adapter transforms the extracted features before passing them to the classifier head g_{ω} . The classification head weight matrix is denoted by $\mathbf{W}_{\omega} \in \mathbb{R}^{d \times C}$, where C represents the number of classes to be classified. The adapter and classifier-combined network produces a C -dimensional logit vector, which is converted into a probability distribution $\mathbf{p}$ over the C classes using the softmax function [45]. The summarized process is given in Eq. (2):

$$\mathbf{p} = \frac{\exp(g_{\omega}(A_{\phi}(\mathbf{x}))_c / \tau)}{\sum_{c'=1}^C \exp(g_{\omega}(A_{\phi}(\mathbf{x}))_{c'} / \tau)} \quad (2)$$

where τ is the temperature parameter controlling the softness of the probability vector and $\mathbf{p}$ denotes the probability vector, indicating the likelihood that the image vector $\mathbf{x}$ belongs to one of the C classes. Specifically, $g_{\omega}(A_{\phi}(\mathbf{x}))_c$ represents the logit corresponding to class c .

The cross-entropy loss [45] for a single sample $\mathbf{x}$ with predicted probability vector $\mathbf{p}$ in the multiclass classification scenario is defined as follows:

$$L_{\text{CE}} = - \sum_{c=1}^C y_c \log(p_c) \quad (3)$$

where y_c represents a one-hot encoded representation for class c , p_c denotes the predicted probability that the sample belongs to class c , and C indicates the total number of classes.

We minimized the cross-entropy loss on randomly sampled targets T from D_{ft} to optimize model performance, considering the predictions made by the classifier g_ω based on the features extracted by f_θ and adapted by A_ϕ . The objective function is formulated as follows:

$$\mathcal{O}_{\text{ft}} = \arg \min_{\phi, \omega} \sum_{i=1}^T L_{\text{CE}}(g_\omega(A_\phi(\mathbf{x})), \mathbf{y}) \quad (4)$$

where $\mathcal{O}_{\text{ft}}$ represents the optimized objective value to fine-tune the adapter. Fig. 1a illustrates the fine-tuning mechanism of the model.

1) RESIDUAL CONNECTION AND FEATURE NORMALIZATION

We further adopted residual connections to blend the fine-tuned features dynamically with the original features and applied feature normalization to mitigate overfitting and improve the robustness of the adapters. Fig. 1a depicts this process.

$$\mathbf{x}^* = \frac{\mathbf{x} + \mathbf{x}_a}{\|\mathbf{x} + \mathbf{x}_a\|_v} \quad (5)$$

where $\mathbf{x}$ denotes the high-dimensional feature vector processed by the frozen encoder f_θ , $\mathbf{x}_a$ represents the feature transformed by the adapter $A_\phi(\cdot)$, and $\mathbf{x}^*$ indicates the normalized feature vector. The denominator can be expressed as follows:

$$\|\mathbf{x} + \mathbf{x}_a\|_v = \left(\sum_{j=1}^d |x_j|^v \right)^{1/v} \quad (6)$$

where d represents the feature dimension, v denotes the norm parameter (e.g., $v = 2$ for the Euclidean norm), and x_j represents the j th element of the feature vector $(\mathbf{x} + \mathbf{x}_a)$ [46]. Eq. (6) defines the denominator term in feature normalization $\|\mathbf{x} + \mathbf{x}_a\|_v$, clarifying that it is the sum of the absolute values of each element raised to the power v , and the result is raised to the power of $\frac{1}{v}$.

C. FEW-SHOT LEARNING

This study applies an N -way K -shot framework to evaluate the ability of the model to generalize from a limited number of examples. The query set $\mathcal{Q}$ comprises additional samples from the same classes. This fine-tuning procedure simulates real-world scenarios where labeled data is scarce, challenging the model to learn effectively with minimal supervision. During model evaluation, we detached the classification head $g_\omega(\cdot)$, and the logits were obtained from $f_\theta(A_\phi(\cdot))$, where $f_\theta(\cdot)$ remained frozen. Fig. 1 and Algorithm 1 provide an overview of how prototypical FSL is incorporated into this study. This unified procedure addresses the primary objective of enabling robust classification under scarce annotations by dynamically fine-tuning the adapter while preserving pretrained knowledge in $f_\theta(\cdot)$.

Algorithm 1 Adapter-Based Histopathology Classification With Few-Shot Learning

Require: D , $f_\theta(\cdot)$, $A_\phi(\cdot)$, $g_\omega(\cdot)$, Few-shot algorithms: ProtoNet, SimpleShot, BDCSPN, Total epochs: E .

Ensure: Freeze f_θ

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1: Dataset preprocessing: Divide  $D$  into  $D_{\text{ft}}$  (fine-tune),
    $D_{\text{val}}$  (validation), and  $D_{\text{test}}$  (testing)
2: for  $epoch = 1$  to  $E$  do
3:   Sample batch  $\{(\mathbf{I}_i, y_i)\}$  from  $D_{\text{ft}}$ 
4:    $\mathbf{x}_i \leftarrow f_\theta(\mathbf{I}_i)$  ▷ Extract features
5:    $\mathbf{x}_a \leftarrow A_\phi(\mathbf{x}_i)$  ▷ Adapter output
6:    $\mathbf{x}^* \leftarrow \frac{\mathbf{x}_i + \mathbf{x}_a}{\|\mathbf{x}_i + \mathbf{x}_a\|_v}$  ▷ Apply residual + normalization
7:    $\mathbf{p}_i \leftarrow g_\omega(\mathbf{x}^*)$  ▷ Compute logits
8:    $L_{\text{CE}} = -\sum y_i \log(\mathbf{p}_i)$  ▷ Training loss
9:   Update  $\phi, \omega$  via backpropagation
10:  Freeze  $A_\phi, g_\omega$ 
11:  Construct  $N$ -way  $K$ -shot tasks from  $D_{\text{val}}$  with  $\mathcal{S}, \mathcal{Q}$ 
12:  for FSL algorithms do
13:    Compute class prototypes  $\mu_c$ .
14:    Classify query  $\mathcal{Q}$  samples.
15:    Record few-shot validation accuracy.
16:  end for
17:  Track best validation performance
18: end for

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Note: $\{\text{ProtoNet}, \text{SimpleShot}, \text{BDCSPN}\} \in \text{FSL algorithms}$.

Various prototypical FSL methods are applied in this work [37], [40], [41]. Next, this study provides an overview of the algorithms.

ProtoNet: The ProtoNet [37] computes a prototype (mean embedding) for each class in the support set and classifies query samples based on their similarity to these prototypes. The mathematical formula to compute the mean prototype vector μ_c for each class c in the support set $\mathcal{S}_c$ is given by the following:

$$\mu_c = \frac{1}{|\mathcal{S}_c|} \sum_{(\mathbf{x}_i^s, y_i^s) \in \mathcal{S}_c} A_\phi(\mathbf{x}_i^s) \quad (7)$$

where $\mathcal{S}_c$ denotes the set of support samples labeled class c , and A_ϕ represents the adapter network. The probability $p(y = c|\mathbf{x}_j^q)$ of assigning the query sample $\mathbf{x}_j^q$ to class c is calculated using the Euclidean distance $\delta(\cdot, \cdot)$ as follows:

$$p(y = c|\mathbf{x}_j^q) = \frac{\exp(-\delta(A_\phi(\mathbf{x}_j^q), \mu_c))}{\sum_{c'=1}^C \exp(-\delta(A_\phi(\mathbf{x}_j^q), \mu_{c'}))}. \quad (8)$$

The class prototypes computed using Eq. (7) can classify each query sample $\mathbf{x}_j^q$ by finding the nearest prototype using Eq. (8).

SimpleShot Algorithm: SimpleShot [40] improves the ProtoNet [37] by introducing feature normalization before calculating the prototype vectors using Eq. (7), and it applies cosine similarity instead of the Euclidean distance to classify query examples. The probability $p(y = c|\mathbf{x}_j^q)$ of assigning the query sample $\mathbf{x}_j^q$ to the c th class is calculated using cosine

similarity in the SimpleShot algorithm [40] as follows:

$$p(y = c | \mathbf{x}_j^q) = \frac{\exp(\cos(A_\phi(\mathbf{x}_j^q), \boldsymbol{\mu}_c))}{\sum_{c'=1}^C \exp(\cos(A_\phi(\mathbf{x}_j^q), \boldsymbol{\mu}_{c'}))}. \quad (9)$$

The normalized class prototypes calculated using Eq. (7) can classify each query sample $\mathbf{x}_j^q$ based on cosine similarity using Eq. (9).

BDCSPN Algorithm: Bias-diminishing cosine similarity-based prototypical network (BDCSPN) [41] addresses intra-class and cross-class bias in prototype estimation. Like SimpleShot [40], this approach computes basic prototypes for each class c and fine-tunes the network $A_\phi(\mathbf{x}_i^s, \mathbf{y}_i^s)$ using a cosine similarity-based classifier on the base classes as presented in Eq. (9).

This method is unique due to its refinement of the initial prototypes with a pseudo-labeled support set to minimize intraclass bias, which refers to the difference between the computed and expected prototypes. The refinement process involves a subset of query samples confidently classified for each class, which are applied to adjust the initial prototypes $\boldsymbol{\mu}_c$ (computed using Eq. (7)), resulting in refined prototypes $\hat{\boldsymbol{\mu}}_c$ [41]. Additionally, the BDCSPN addresses cross-class bias by adjusting the normalized query features. This adjustment involves shifting the query features toward support features [41]. Finally, query samples $\mathcal{Q}$ are classified using the refined prototypes $\hat{\boldsymbol{\mu}}_c$ and adjusted query features $\mathcal{Q}$ with the cosine similarity function in Eq. (9) [41].

D. EVALUATION PROTOCOLS

The general evaluation protocol is to pass the patch images through the network $g_\omega(A_\phi(f_\theta(\cdot)))$, as depicted in Fig. 1a, and evaluate performance. Furthermore, we evaluated model performance using linear probes and by classifying the high-dimensional features from the adapter networks. These alternative evaluation protocols are discussed in this section.

1) LINEAR PROBE EVALUATION

For the linear probe evaluation, the adapter head was removed, and patch classification tasks were performed using features obtained via Eq. (1) from the frozen foundational model encoders $f_\theta(\cdot)$. The input patches were resized to match the required input shape of the ViT-B [43] or ResNet50 [25] encoders. The linear evaluation was conducted by fine-tuning the linear classification head $g_\omega(\cdot)$ using the softmax activation function [45] and the cross-entropy loss defined in Eq. (3) that mapped the high-dimensional encoder features to the dataset classes (Fig. 1c). The softmax activation function for the logits obtained from $g_\omega(\cdot)$ is as follows:

$$\sigma(g_\omega(\mathbf{x}_i)) = \frac{\exp(g_\omega(\mathbf{x}_i))}{\sum_{j=1}^C \exp(g_\omega(\mathbf{x}_j))} \quad (10)$$

where $\sigma(g_\omega(\mathbf{x}_i))$ represents the probability that input $\mathbf{x}_i$ belongs to class i , $g_\omega(\mathbf{x}_i)$ indicates the logit for input $\mathbf{x}_i$, and C denotes the total classes.

2) PROTOTYPE-BASED EVALUATION

Inspired by the nearest centroid classifier [47], [48], we designed an evaluation protocol where class labels are assigned to the instances based on the proximity of their feature representations to the mean vectors or prototypes of each class in the feature space following Eq. (7). We adopted this widely used protocol from prior literatures [17], [18], [37], [40], and [49]. During the evaluation, we detached the classification head $g_\omega(\cdot)$ and obtained the logits from $A_\phi(f_\theta(\cdot))$, where $f_\theta(\cdot)$ remains frozen. When presented with new testing data $\mathcal{D}_{test}$, the adapter predicts the class labels by identifying the closest mean feature vector or prototype using Eq. (7). Fig. 1b depicts this process. Following the FSL strategy to fine-tune the adapter modules, we employed the prototype assignment mechanism to evaluate the performance on the testing set using high-dimensional adapter features, assessing model performance.

IV. EXPERIMENT SETUP

A. DATASETS

Inspired by recent experiments conducted by Peng Gao et al. [29], we randomly sampled a limited number of instances per class, specifically 1, 2, 5, 10, 15, and 20 examples, to fine-tune the adapter and enable comparative analyses. Table 2 provides a detailed overview of several datasets focusing on colorectal, lung, and breast tissue, including dataset specifications, class diversity, image sizes, and data origins for medical image analysis in this study.

The NCT [50] and Kather [51] datasets contain multiclass colorectal cancer datasets, whereas the LC25000 [52] dataset contains images of tissue patches from colon and lung cancer. The NCT dataset comprises 100,000 image patches for training and validation, and 7180 for testing. The training dataset was further divided into an 80:20 split for fine-tuning and validation, whereas the 7180 images were used exclusively for testing. For the Kather and LC25000 datasets, we partitioned the data into 60% for randomly sampling a few examples to fine-tune the network, 20% for few-shot validation, and 20% for testing. The Camelyon17 Wilds [53] and PCAM [54] datasets are aimed at classifying tumor tissue in breast cancer histopathology images, consisting of predefined splits comprising 80% for training, 10% for validation, and 10% for testing. Table 2 presents the details of these datasets.

The PCAM [54] and Camelyon17 Wilds [53] datasets are derived from the Camelyon16 [55] and Camelyon17 [56] challenges, respectively. The PCAM dataset represents a straightforward patch classification task, whereas Camelyon17 Wilds presents additional challenges due to data variations across hospitals, as its training, validation, and testing sets were curated from distinct institutions [53]. Given these considerations, we conducted in-depth experiments using the Camelyon17 Wilds dataset to evaluate the robustness of the proposed approach under complex, domain-aware conditions. Furthermore, to ensure a comprehensive

TABLE 2. Downstream pathology datasets details for foundation models evaluation.

Organ	Colorectal	Colorectal	Lung and colon	Breast	Breast
Dataset	NCT [50]	Kather [51]	LC25000 [52]	Camelyon17 Wilds [53]	PCAM [54]
No. of images	1,07,180	5,000	25,000	4,55,954	3,27,680
No. of classes	9	8	5	2	2
Class name	Adipose, background, debris, lymphocyte, mucus, muscle, normal, stroma, tumor	Adipose, complex, debris, empty, lymphocyte, mucosa, stroma, tumor	Benign colon tissue, colon adenocarcinoma, benign lung tissue, lung adenocarcinoma, lung squamous cell carcinoma	Tumor, normal	Tumor, normal
Patch size	224 × 224	150 × 150	768 × 768	96 × 96	96 × 96
Data origin	Heidelberg University	Heidelberg University	James A. Haley Veterans' Hospital	Radboud University	Radboud University

Note: The PCAM dataset was used for the state-of-the-art comparison.

TABLE 3. Percentage of images per class used to fine-tune the adapter.

Dataset	No. of classes	Images per class					
		1	2	5	10	15	20
NCT [50]	9	0.008%	0.017%	0.042%	0.084%	0.126%	0.168%
Kather [51]	8	0.16%	0.32%	0.80%	1.60%	2.40%	3.20%
LC25000 [52]	5	0.020%	0.040%	0.100%	0.20%	0.30%	0.40%
Camelyon17 Wilds [53]	2	0.0004%	0.0008%	0.0022%	0.0044%	0.0066%	0.0088%

evaluation, we validated the method against state-of-the-art (SOTA) methods [29], [30], [57], [58], [59], [60] in a dataset-constrained scenario using the PCAM dataset. We employed 0.5% of the available data for training to perform SOTA comparisons, consistent with the experimental framework reported in previous work [30].

Similar settings, specifically 1, 2, 5, 10, 15, and 20 examples per class, were applied for the linear probe experiments. Table 3 presents the varying number of classes across datasets, which led to fine-tuning the adapter networks on different sample sizes. For instance, in the one-image-per-class setting, for the NCT [50] dataset, the network had access to nine images when fine-tuning the adapter module. Kather [51] used eight images, LC25000 [52] used five, and Camelyon17 Wilds [53] used two images in the same settings. Table 3 presents a detailed overview of the percentages of histopathology image patches used to fine-tune the adapters. We ensured that no overlap occurred between the classes in the training, validation, and testing sets to assess the generalizability of the model. The datasets span various organs and institutions, introducing variability in the tissue types and imaging conditions, which is critical for evaluating the robustness of the proposed approach.

B. HISTOPATHOLOGY PRETRAINED FOUNDATION MODELS

The publicly available pretrained histopathology foundation models in this study used as feature encoders include

Barlow-twins (BT) [19], MoCo-v2 (MoCo) [19], SwAV [19], and iBOT [12]. Table 4 provides a detailed overview of these pretrained histopathology foundation models.

TABLE 4. Details of pretrained foundation models used as encoders.

Pretraining method	Backbone	No. of params	Dataset	Patches (M)	Training epochs
Barlow-twins (BT) [19]	ResNet50	23.5 M	TCGA	19	200
MoCo-v2 (MoCo) [19]	ResNet50	23.5 M	TCGA	19	200
SwAV [19]	ResNet50	23.5 M	TCGA	19	200
iBOT [12]	ViT-B	85.8 M	PanCancer	43.3	-

Note: M: millions.

The BT [10], MoCo [11], and SwAV [9] models obtained from Kang et al. [19] were pretrained on 19 million image patches from the TCGA cohort for 200 epochs. The BT, MoCo, and SwAV models were pretrained on images with 512×512 pixels with multiple fields of view at $20\times$ and $40\times$ magnification, employing the ResNet50 [25] backbone consisting of approximately 25.6 million parameters and resulting in a final feature representation of 2048 dimensions. In contrast, the iBOT model was pretrained on the PanCancer cohorts comprising 43.3 million image patches for 155,000 iterations and employed a ViT-B [43] backbone with a pretraining image size of 224×224 pixels, comprising about 85.8 million parameters and an output dimension of 768 [12].

The naming convention for other models in this study was applied to the Phikon model introduced by Filiot et al. [12], which was referred to as iBOT. This approach ensures clarity and coherence when comparing results across models.

C. EXPERIMENT PARAMETERS

All experiments were conducted using the PyTorch-2.0 framework [61]. During the fine-tuning phase, the foundation model encoder $f_\theta(\cdot)$ was frozen and used as a feature extractor. The adapter $A_\phi(\cdot)$ and classification head $g_\omega(\cdot)$ were fine-tuned on a few examples per class to minimize the cross-entropy loss between the predicted and actual class labels, as defined by the principles outlined in Eq. (4), where the AdamW optimizer [62] was employed with an initial learning rate of 0.0001 and was fine-tuned for 100 epochs with a cosine annealing learning rate scheduler [63] to optimize the objective function. Once the adapter $A_\phi(\cdot)$ was tuned, the model was validated on a series of few-shot tasks. Similar settings were employed in the linear probe experiments.

We followed a standard N -way K -shot protocol during the few-shot validation experiments, similar to the settings widely applied in many prior studies [13], [44] [37], [40], [33], [41]. The variable N represents the number of classes, and K denotes the number of support samples per class. We modified the experimental strategy from prior work [29], [44] by employing adapters to enable foundation models for visual classification instead of using adapters to align multimodal visual and textual data. Detailed fine-tuning and evaluation strategies were discussed in the method section. We used similar settings when evaluating this work against the SOTA methods, with the only difference being the number of samples per class (700 in this case) when experimenting on the PCAM dataset, resulting in approximately 0.5% dataset utilization.

We also calculated the computational costs following the protocol from Pucci et al. [64] by performing 100 repeated forward passes on dummy inputs with a batch size of 256 (default size $1 \times 3 \times 224 \times 224$). The throughput is calculated as the number of samples processed per second, and Table 8 reports the peak memory usage in megabytes.

D. QUALITATIVE ANALYSIS ON WHOLE SLIDE IMAGES

We performed additional qualitative analyses to assess the generalizability of the proposed method. Multiple breast cancer whole slide images (WSI) from the Camelyon dataset [55] were employed to assess model performance on binary classification tasks. The best-performing model was selected, consisting of the pretrained iBOT encoder with an MLP adapter, and fine-tuned via the BDCSPN algorithm on the WCAM dataset using twenty images per class. The WSIs were partitioned into patch sizes of 224×224 pixels at $40\times$ magnification. The model then predicted whether each patch contained tumor tissue or not, highlighting the qualitative performance of the proposed approach.

V. RESULTS

A. ADAPTER PERFORMANCE ACROSS DATASETS AND FEW-SHOT ALGORITHMS

Fig. 2 illustrates the critical findings from the analysis of the MLP-based adapters fine-tuned using FSL algorithms across datasets and foundation models. The results were analyzed as the number of images per class increased from 1 to 20. The supplementary material (Fig. S1) presents a detailed analysis of the Conv adapters.

The adapter classification performance was evaluated using the method outlined in Section III-D2. The results were compared to the linear probe approach described in Section III-D2. These comparative analyses, presented as Fig. 2 and Fig. S1, provide insight into the capacity of the proposed method to learn downstream classification tasks and achieve high classification accuracy. The models exhibited comparable performance in both cases. The MLP adapters outperformed the Conv adapters; hence, the MLP adapter results were included in the main text.

1) FOUNDATION MODEL SENSITIVITY

The sensitivity of the foundation models (BT, iBOT, MoCo, and SwAV) to the number of available data was significant, particularly in low-data regimes (Fig. 2a). The iBOT and BT models consistently outperformed the others when the number of images per class was limited. For instance, with just one image per class, iBOT reached about 0.82 accuracy on the NCT and LC25000 datasets, whereas MoCo and SwAV lagged at 0.77 and 0.75 accuracy, respectively. As the number of images increased to 20, the performance gap diminished, and all models converged to more than 91% accuracy. However, in data-constrained scenarios (one to five images per class), iBOT and BT demonstrated superior robustness in classification tasks, outperforming their counterparts.

2) PERFORMANCE ACROSS DATASETS

The performance of MLP-based adapter fine-tuning varies drastically across histopathology datasets, as depicted in Fig. 2a. The Camelyon17 Wilds dataset posed the most significant challenge, with performance starting at 0.64 with one image per class and only improving to 0.90 with 20 images per class. In contrast, NCT and LC25000 exhibited smoother and higher initial performance, starting near 0.80 and reaching over 0.93 as more images became available. This result reflects the well-separated nature of the tissue classes in these multiclass datasets, offering a less complex classification task than Camelyon17 Wilds. Another noticeable trend from Fig. 2a and Fig. 2b is that the performance increases considerably with an increasing number of images per class across all datasets.

3) ALGORITHM COMPARISONS ACROSS DATASETS AND MODELS

The performance of FSL algorithms (BDCSPN, SimpleShot, and ProtoNet) varied significantly across datasets and

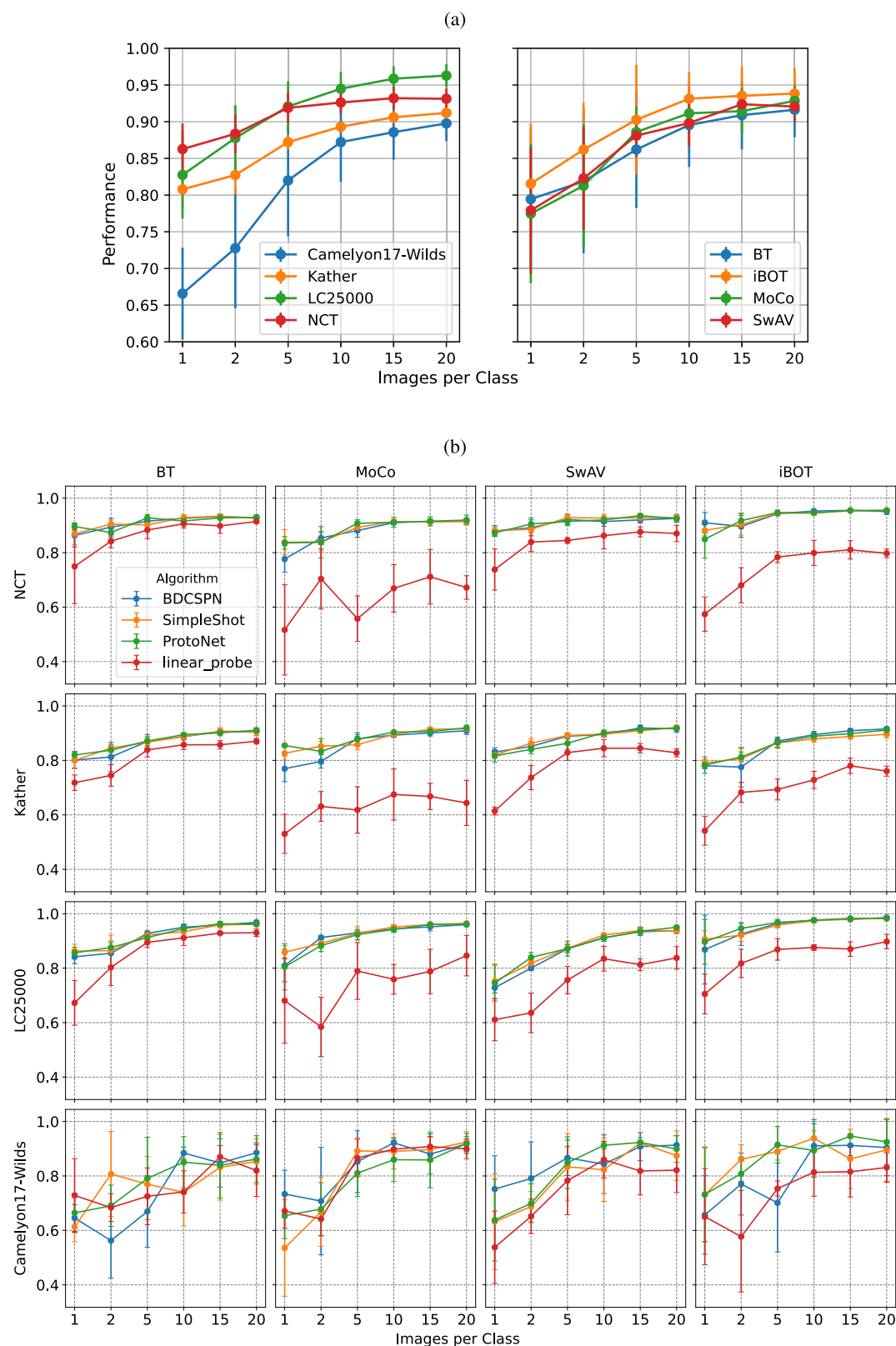


FIGURE 2. Histopathology classification performance of the multilayer perception (MLP) adapter. (a) Average classification accuracy of MLP adapters across datasets (left) and models (right). (b) Detailed average performance of MLP adapters when fine-tuned with various algorithms across all datasets.

foundation models (BT, iBOT, MoCo, and SwAV), as illustrated in Fig. 2b. The BDCSPN algorithm achieved the best performance across NCT and LC25000, with accuracy reaching 0.96 and 0.97, respectively, when fine-tuned with iBOT. SimpleShot followed closely but still trailed by a margin of 2% to 3% compared to BDCSPN. This trend held across models, with iBOT paired with BDCSPN performing better on NCT and Kather than BT, MoCo, and SwAV. On NCT, iBOT with BDCSPN reached an accuracy of 0.96, and BT with BDCSPN achieved 0.94 accuracy.

The ProtoNet displayed reasonable performance but consistently underperformed compared with the BDCSPN and SimpleShot few-shot algorithms. For example, on LC25000, ProtoNet achieved 0.92 accuracy, which is significantly less than 0.97 for BDCSPN. This gap was observed across the foundation models as well, with iBOT outperforming other models even when paired with the less effective ProtoNet. On Camelyon17 Wilds, ProtoNet with iBOT reached 0.968 accuracy, which was lower than iBOT with BDCSPN, although higher than most other models using ProtoNet. The linear probe method consistently performed the worst across datasets and models. With NCT and Camelyon17 Wilds, the linear probe achieved as low as 0.50 to 0.65 accuracy with one image per class and only improved to 0.70 to 0.75 with 20 images. This performance was consistent across all foundation models, with slight variances between iBOT, BT, MoCo, and SwAV, indicating the limitation of the linear probe method in adapting to various backbones and datasets, especially in FSL scenarios.

Overall, BDCSPN and SimpleShot outperformed ProtoNet and the linear probe methods, and iBOT consistently achieved higher accuracy than other foundation models across all algorithms.

4) QUANTITATIVE COMPARISON WITH SOTA METHODS TESTED ON PCAM DATASET

We selected the ResNet50-based SwAV model fine-tuned with 700 examples per class by applying the SimpleShot algorithm, resulting in using approximately 0.5% of the available data during the entire fine-tuning process for a fair comparison (see Section IV-C). The results in Table 5 demonstrate that the proposed method significantly outperforms the competing SOTA methods [29], [30], [57], [58], [59], [60]. The compared SOTA methods were developed to address data-constrained scenarios using a ResNet50 backbone, as reported in previous work [30]. The MLP and Conv adapter approaches performed better than the SOTA methods. In contrast, the semi-supervised and multimodal CLIP methods tested on PCAM yielded lower performance. The superior performance of the proposed method underscores its strong generalizability while reinforcing its potential for real-world histopathological analysis.

B. COMPARATIVE ANALYSIS OF MLP VS. CONV ADAPTERS

This section discusses the results of the comparison between the MLP and Conv adapters. The focus is on performance

across datasets, backbone sensitivity, and how the models handle increased data.

1) COMPARISON OF MLP AND CONV ADAPTERS ACROSS DATASETS

The MLP adapters consistently outperformed the Conv adapters across multiple datasets, particularly in low-data regimes (Fig. 3a). In LC25000, the MLP adapters achieved an accuracy of 0.96 with 20 images per class, compared to 0.94 for Conv adapters. A similar trend was observed in other datasets, where MLP adapters demonstrated increased adaptability. For instance, in Camelyon17 Wilds, MLP adapters reached 0.83 with 20 images per class, significantly surpassing Conv adapters, which started at 0.64 with one image per class and only improved marginally with more data. This performance gap was even more apparent in data-constrained scenarios, underscoring the efficiency of the MLP adapters in such settings. Another trend observed in Fig. 2b and Fig. S1b is the high standard deviation of all models when the number of images per class was one or two.

2) PERFORMANCE VARIABILITY AND DISTRIBUTION

The performance distributions based on classification accuracy also highlighted the consistency of the MLP adapters over Conv adapters. As observed in the box plots (Fig. 3b), MLP adapters displayed tighter performance distributions, with an interquartile range of 0.04 in the NCT dataset, compared to 0.08 for Conv adapters. The reduced variability made MLP adapters more reliable across datasets, whereas Conv adapters exhibited higher variability, particularly in Camelyon17 Wilds.

3) ADAPTER SENSITIVITY TO BACKBONE ARCHITECTURES

Both MLP and Conv adapters displayed sensitivity to various backbone architectures, but MLP adapters consistently maintained superior performance across encoders, such as BT, MoCo, and iBOT (Fig. 3c). For example, with the iBOT encoder, the MLP adapters maintained an accuracy range of 0.92 to 0.96, whereas the Conv adapters exhibited a broader, lower spread, particularly with MoCo. The Conv adapters exhibited a broader classification performance spread across backbones, ranging from 0.75 to 0.90 and averaging 0.83, compared to the more consistent performance accuracy of the MLP adapters, which clustered near 0.90 to 0.94.

Overall, the MLP adapters demonstrated greater robustness and reliability for few-shot adaptation tasks in data-constrained scenarios.

C. ABLATION STUDIES

1) HIDDEN DIMENSIONS OF THE MLP ADAPTER BOTTLENECK LAYERS

Following the work by Gao et al. [29], we selected 512 as the hidden dimensionality for the bottleneck layers, corresponding to approximately one-quarter of the input dimension because it is consistent with the design principles of the

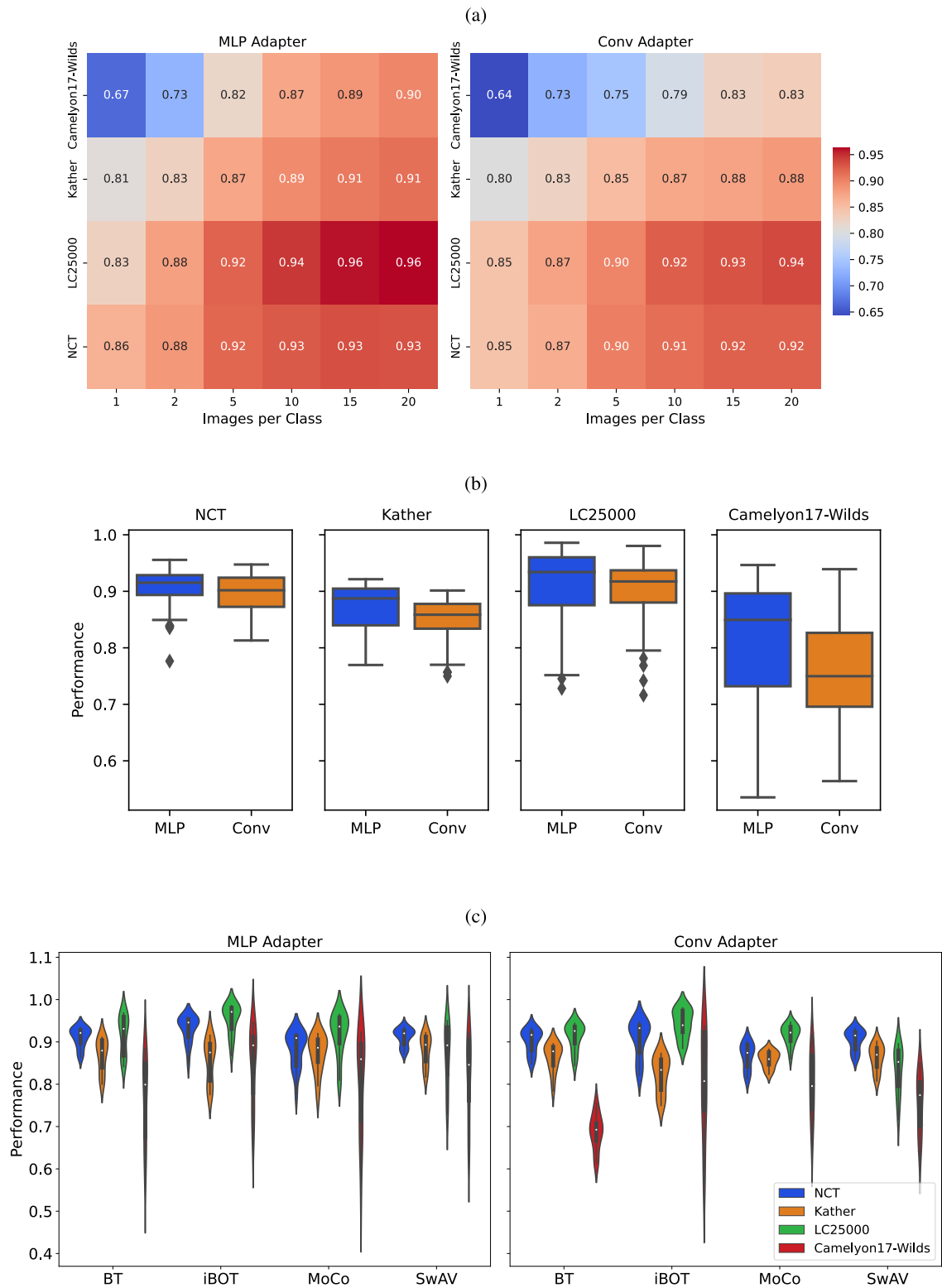


FIGURE 3. Performance comparison between multilayer perception (MLP) and convolutional (Conv) adapters. (a) Average classification accuracy of adapters with an increasing number of image patches across datasets. (b) Box plot depicting the MLP vs. Conv adapter trends on the datasets. (c) Violin plot comparing the Conv and MLP adapter classification accuracy using various backbones and on the datasets.

TABLE 5. Quantitative performance comparison with State-of-the-art (SOTA) methods on the PCAM dataset with limited data usage.

Learning strategy	SOTA methods	Data usage	Average accuracy	Precision	Recall	F1-score	AUC
Semi supervised	DASH [57]	0.5%	0.71 ($\pm$ 0.98)	0.753 ($\pm$ 0.78)	0.702 ($\pm$ 0.45)	0.727 ($\pm$ 0.50)	0.82 ($\pm$ 0.02)
Semi supervised	FixMatch [58]	0.5%	0.732 ($\pm$ 0.76)	0.775 ($\pm$ 0.50)	0.737 ($\pm$ 0.25)	0.756 ($\pm$ 0.37)	0.84 ($\pm$ 0.04)
Semi supervised	FlexMatch [59]	0.5%	0.74 ($\pm$ 0.80)	0.781 ($\pm$ 1.15)	0.73 ($\pm$ 0.90)	0.755 ($\pm$ 0.84)	0.85 ($\pm$ 0.03)
Multimodal (CLIP)	CoOp [60]	0.5%	0.636 ($\pm$ 0.25)	0.639 ($\pm$ 0.42)	0.625 ($\pm$ 0.35)	0.630 ($\pm$ 0.29)	0.67 ($\pm$ 0.02)
Multimodal (CLIP)	CLIP-Adapter [29]	0.5%	0.723 ($\pm$ 1.02)	0.772 ($\pm$ 0.95)	0.632 ($\pm$ 0.56)	0.694 ($\pm$ 0.84)	0.81 ($\pm$ 0.02)
Multimodal (CLIP)	CLIPath [30]	0.5%	0.815 ($\pm$ 0.78)	0.794 ($\pm$ 0.35)	0.85 ($\pm$ 0.80)	0.821 ($\pm$ 0.95)	0.89 ($\pm$ 0.02)
Proposed method	MLP adapter	0.5%	0.861 ($\pm$ 0.005)	0.866 ($\pm$ 0.005)	0.861 ($\pm$ 0.005)	0.861 ($\pm$ 0.005)	0.931 ($\pm$ 0.003)
Proposed method	Conv adapter	0.5%	0.859 ($\pm$ 0.009)	0.863 ($\pm$ 0.008)	0.859 ($\pm$ 0.009)	0.858 ($\pm$ 0.01)	0.941 ($\pm$ 0.004)

ResNet50 architecture used in the three models obtained from Kang et al. [19]. We conducted a limited comparative experiment using 192 and 512 hidden dimensions, focusing solely on the NCT dataset and fine-tuning the MLP adapters by employing the BDCSPN FSL algorithm and using the ViT-based iBOT model obtained from Filot et al. [12] with a 768-dimensional output embedding from the encoder to justify using the same hidden dimensions in the iBOT model. Table 6 presents the quantitative classification performance of the MLP adapters with various hidden layers using the iBOT model.

Although this experiment was constrained by its application to a single dataset, the NCT dataset was chosen due to its robust classification performance, featuring nine well-separated classes. The high classification accuracy in this dataset provided a reliable benchmark for evaluating the potential influence of various hidden dimensions on model performance. As presented in Table 6, the iBOT model performed comparably well when using 512-dimensional hidden layers.

However, this experimental scope was limited because it did not account for the full diversity of the available datasets, such as those with two, five, or eight classes, presenting different levels of classification difficulty. Despite this limitation, the choice of a 512-dimensional hidden layer for the MLP adapter in the ViT-based iBOT model is justified by the need to maintain architectural consistency across models and the positive performance observed in the NCT dataset. The decision to standardize the hidden dimensions to 512 across models also simplifies the model performance comparison across encoder/backbone architectures, ensuring that any observed differences are more likely to be due to the model characteristics rather than variations in the adapter configuration.

Although based on limited experimentation, this approach offers a practical balance between performance consistency and architectural coherence, justifying the selection of 512 as the hidden dimensions.

2) RESIDUAL CONNECTIONS IN MLP ADAPTERS

The NCT is a well-balanced, multiclass dataset with the highest number of classes in this study. Therefore, we selected

the average performance of the SimpleShot model with the highest average accuracy and area under the curve (AUC) metrics, presented in Table S2 in the supplementary material. We inspected the visual landscape of the NCT dataset and how the performance improves with the adapter method compared to the linear probe. Table 7 presents the results of fine-tuning the MLP adapter using the SimpleShot algorithm, whereas Fig. 4 illustrates how the model learns the manifolds.

We visualized the high-dimensional feature space and projected the predicted classification outputs using the *t*-stochastic neighbor embedding (*t*-SNE) [65] from the Scikit-learn library [66]. Fig. 4 illustrates that the linear probe method struggles to distinguish between classes in the low-data regime. In contrast, fine-tuning the adapter with the SimpleShot algorithm enables accurate differentiation between classes. Additionally, Fig. 4 demonstrates that models incorporating the residual connections more clearly separate the tumor areas (gray) from the stroma (pink) and muscle (yellow) regions compared to models without residual connections. Similarly, Table 7 confirms improved model performance when residual connections are employed.

3) COMPUTATIONAL TRADE-OFFS

Table 8 summarizes the averaged mean inference time, standard deviation, throughput, memory usage, and model parameter counts for the adapter networks based on ResNet50 and ViT-B. We averaged the inference metrics using a dummy input batch size of 256, where each input is 224×224 pixels for each backbone and adapter type to provide an overview of the computational costs of the proposed few-shot adaptation framework. The detailed experimental procedure is discussed in IV-C. The ResNet50-based MLP adapters process inputs faster, yielding high throughput while requiring 130.78 MB of memory. The ViT-based models require more time to process the inputs with a lower throughput rate, despite using less GPU memory overall to process the inputs.

Notably, the ResNet50-based MLP adapters process inputs faster and yield higher throughput, making them advantageous for resource-constrained or latency-sensitive

TABLE 6. Quantitative classification performance of the multilayer perceptron (MLP) adapters with hidden layers using the iBOT model.

Images per class	Hidden dimensions	Accuracy	Precision	Recall	F1-score	AUC
1	192	0.389 (± 0.195)	0.581 (± 0.128)	0.358 (± 0.175)	0.293 (± 0.183)	0.828 (± 0.111)
	512	0.786 (± 0.241)	0.867 (± 0.026)	0.735 (± 0.264)	0.724 (± 0.270)	0.958 (± 0.045)
2	192	0.708 (± 0.138)	0.781 (± 0.064)	0.686 (± 0.053)	0.652 (± 0.141)	0.962 (± 0.018)
	512	0.817 (± 0.119)	0.835 (± 0.052)	0.782 (± 0.113)	0.772 (± 0.127)	0.973 (± 0.016)
5	192	0.849 (± 0.065)	0.846 (± 0.038)	0.835 (± 0.053)	0.815 (± 0.062)	0.985 (± 0.004)
	512	0.842 (± 0.021)	0.874 (± 0.093)	0.820 (± 0.223)	0.805 (± 0.249)	0.975 (± 0.038)
10	192	0.906 (± 0.027)	0.887 (± 0.024)	0.879 (± 0.030)	0.873 (± 0.034)	0.989 (± 0.003)
	512	0.952 (± 0.006)	0.928 (± 0.011)	0.931 (± 0.011)	0.928 (± 0.011)	0.995 (± 0.001)
15	192	0.934 (± 0.009)	0.906 (± 0.013)	0.907 (± 0.009)	0.904 (± 0.012)	0.991 (± 0.002)
	512	0.952 (± 0.005)	0.929 (± 0.007)	0.933 (± 0.010)	0.930 (± 0.008)	0.995 (± 0.002)
20	192	0.939 (± 0.004)	0.910 (± 0.006)	0.910 (± 0.004)	0.909 (± 0.004)	0.989 (± 0.002)
	512	0.950 (± 0.011)	0.926 (± 0.015)	0.929 (± 0.014)	0.925 (± 0.015)	0.995 (± 0.001)

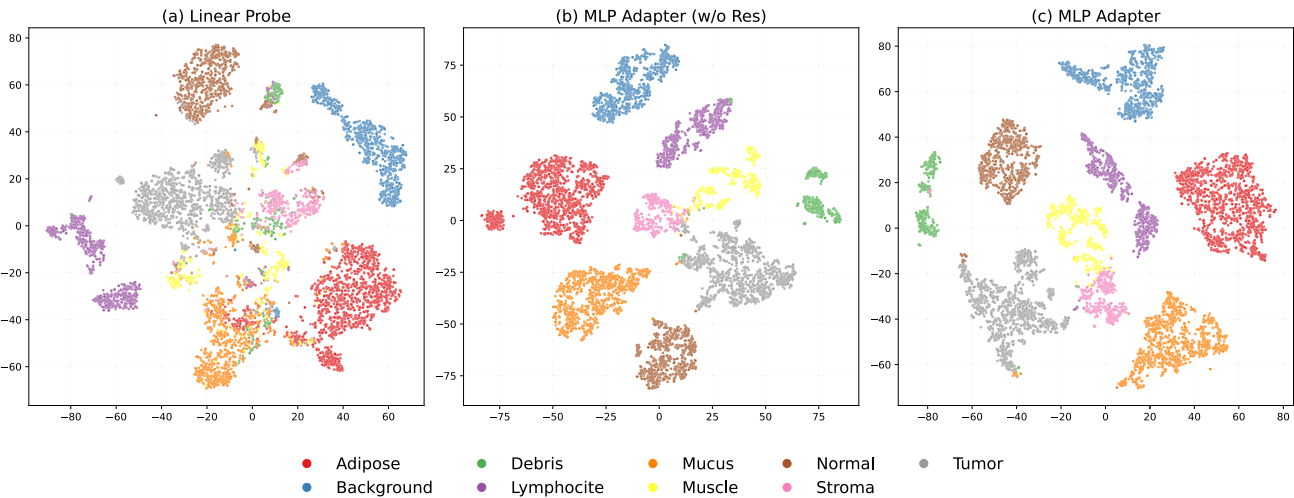


FIGURE 4. Visualization of the learned manifold of the multilayer perceptron (MLP) adapter by fine-tuning the iBOT model using the SimpleShot algorithm.

TABLE 7. Performance summary of iBOT multilayer perceptron (MLP) adapter variants [with and without (w/o) residual connections (res)] with SimpleShot on the NCT dataset, fine-tuned using 20 images per class.

Model	Accuracy	Precision	Recall	F1-score	AUC
Linear probe (no adapter)	0.797 (± 0.015)	0.753 (± 0.028)	0.744 (± 0.023)	0.737 (± 0.021)	0.956 (± 0.008)
MLP adapter (w/o res)	0.939 (± 0.009)	0.909 (± 0.013)	0.914 (± 0.011)	0.908 (± 0.015)	0.990 (± 0.002)
MLP adapter (with res)	0.955 (± 0.002)	0.928 (± 0.004)	0.935 (± 0.003)	0.930 (± 0.003)	0.996 (± 0.0005)

environments. Conversely, ViT-based networks, having a higher number of parameters and slower processing speeds, are better suited for systems that prioritize modeling capability and accuracy over computational efficiency.

TABLE 8. Averaged inference performance and parameter counts across datasets for each backbone and adapter combination.

Backbone + adapter type	No. of params	Mean time (ms)	Throughput (samples/s)	Memory (MB)
ResNet50 + Conv adapter	23.52M	4.53(± 0.18)	56,509	131.48
ResNet50 + MLP adapter	25.62M	4.40(± 0.29)	58,157	130.78
ViT-B + Conv adapter	85.80M	6.31(± 0.30)	40,586	9.64
ViT-B + MLP adapter	86.59M	6.27(± 0.30)	40,772	9.91

Note: M: millions, ms: milliseconds, MB: megabytes.

VI. DISCUSSION

This section presents the interpretation of the findings from the results and explores their broader implications

for FSL in medical image analysis. This section examines the challenges encountered in specific datasets, such as Camelyon17 Wilds, analyzes the implications of the results for medical image classification tasks, and identifies areas for further research and model improvement. Additionally, this section discusses the study limitations and potential future directions to enhance the robustness and scalability of the proposed methods.

A. CHALLENGES IN THE CAMELYON17 WILDS DATASET

The Camelyon17 Wilds dataset presents substantial challenges compared with NCT, Kather, and LC25000, primarily due to its binary classification tasks, high intraclass variability, and domain mismatch due to training, validation, and testing datasets originating from different institutions [53]. Despite only having two classes (tumor vs. normal), the dataset displays significant sample diversity, making generalization more challenging for the models. Starting at a low 0.64 accuracy with one image per class (Fig. 2b), Camelyon17 Wilds gradually improved with more data, reaching 90% accuracy with 20 images per class when using the BDCSPN algorithm and iBOT backbone. However, even with this improvement, the dataset lagged behind NCT and LC25000, which achieved over 93% accuracy. Table 9 presents a detailed comparison of the iBOT model performance when fine-tuned on the Camelyon17 Wilds and NCT datasets to gain a deeper understanding of this performance gap. For this analysis, we examined the detailed average performance in five experiments, where the Conv and MLP adapter-based models were fine-tuned using 20 patch images per class. The table includes the key performance metrics: accuracy, precision, recall, F1-score, and AUC, evaluated using FSL algorithms (i.e., BDCSPN, SimpleShot, and ProtoNet).

Table 9 reveals that the NCT dataset, with its nine distinct tissue classes, provides a more straightforward classification task due to the clear separability between classes (e.g., adipose, lymphocyte, tumor, etc.). The iBOT model consistently achieved high accuracy, often exceeding 0.94, with AUC values surpassing 0.99 when fine-tuning the Conv and MLP adapters with FSL algorithms. The diversity and well-defined nature of the tissue classes in NCT facilitated its robust performance across algorithms, reflecting its ability to generalize well in a multiclass setting. The qualitative results (see Fig. 5 and Fig. S3) further illustrate the efficacy of the proposed method on large-scale WSIs, where the proposed method reliably classifies tumor vs. non-tumor regions on breast cancer WSIs. Section IV-D provides the detailed experimental procedure.

The simple ProtoNet algorithm outperformed more sophisticated models (e.g., SimpleShot and BDCSPN), achieving the highest accuracy of 0.957 with an AUC of 0.985 using the MLP adapter (Table 9). This outcome is likely due

to the comparative limitations of ProtoNet in multiclass separation, which benefit this binary classification task by focusing on finer distinctions between just two classes. Tables S1 and S2 in the supplementary files provide more detailed results for Camelyon17 Wilds and NCT, respectively.

B. IMPLICATIONS FOR MEDICAL IMAGE ANALYSIS

The findings in this work suggest that the iBOT model (particularly with MLP adapters) is well-suited for medical image analysis tasks involving multiclass datasets. However, binary classification tasks (e.g., Camelyon17 Wilds) may require more sophisticated approaches or additional data to achieve comparable performance. The results emphasize the importance of dataset diversity and class separability in model performance and the robustness of MLP adapters in FSL scenarios, primarily when five or more samples are used to fine-tune the adapters. Fig. 5 and Fig. S3 demonstrate that the model failed to classify some regions inside the tumor-annotated areas, potentially indicating misclassified or small non-tumor patches inside the tumor region. However, the substantial overlap between the predicted and annotated tumor regions confirms the effectiveness of the classifier, supporting the results in Table 9. These findings demonstrate the novelty of the proposed approach, especially in the context of large-scale WSIs.

C. LIMITATIONS AND FUTURE WORK

This study is limited in scope, focusing on benchmarking a limited number of datasets and a few types of algorithms with a few selected foundation models. This approach allows for a focused and rigorous analysis of the performance differences between MLP and Conv adapters. However, it constrains the scope of the general feature representation capability of the adapters. The datasets in this study do not encompass the full spectrum of subtypes observed in clinical practice, and highly specialized or rare tumor classes remain underrepresented. The proposed method applies frozen encoders and lightweight adapters, substantially reducing the computational overhead compared with fully fine-tuned backbones. However, scaling to very large backbones or extremely high-resolution histopathology images can still be resource-intensive.

Guo et al. highlighted the benefits of adopting shared foundation models across multiple hospitals, underscoring the challenges of scaling these models to more specialized or rare disease classes [67]. Inspired by this, we recognized the potential of using frozen foundation models hosted on a centralized system, accessible to multiple hospitals via secure devices. Moreover, the way in which lightweight adapters might enable efficient foundation model adaptation to diverse clinical datasets without requiring extensive

TABLE 9. Detailed quantitative performance analysis of Camelyon17 Wilds and NCT datasets using the iBOT model fine-tuned on 20 images per class.

Dataset	Algorithm	Adapter	Accuracy	Precision	Recall	F1-score	AUC
Camelyon17 Wilds	BDCSPN	Conv	0.924 (±0.028)	0.934 (±0.020)	0.924 (±0.028)	0.924 (±0.029)	0.979 (±0.011)
		MLP	0.938 (±0.016)	0.943 (±0.015)	0.938 (±0.016)	0.938 (±0.016)	0.979 (±0.014)
	SimpleShot	Conv	0.923 (±0.025)	0.933 (±0.018)	0.923 (±0.025)	0.922 (±0.025)	0.980 (±0.005)
		MLP	0.877 (±0.141)	0.908 (±0.084)	0.877 (±0.141)	0.865 (±0.166)	0.955 (±0.069)
	ProtoNet	Conv	0.937 (±0.011)	0.943 (±0.009)	0.937 (±0.011)	0.937 (±0.011)	0.979 (±0.006)
		MLP	0.957 (±0.01)	0.958 (±0.01)	0.957 (±0.01)	0.957 (±0.01)	0.985 (±0.007)
NCT	BDCSPN	Conv	0.946 (±0.005)	0.920 (±0.008)	0.926 (±0.008)	0.922 (±0.008)	0.993 (±0.002)
		MLP	0.950 (±0.011)	0.926 (±0.015)	0.929 (±0.014)	0.925 (±0.015)	0.995 (±0.001)
	SimpleShot	Conv	0.949 (±0.003)	0.923 (±0.006)	0.929 (±0.004)	0.925 (±0.005)	0.995 (±0.0007)
		MLP	0.955 (±0.002)	0.928 (±0.004)	0.935 (±0.003)	0.930 (±0.003)	0.996 (±0.0005)
	ProtoNet	Conv	0.943 (±0.005)	0.919 (±0.005)	0.922 (±0.004)	0.920 (±0.004)	0.994 (±0.001)
		MLP	0.954 (±0.004)	0.928 (±0.006)	0.933 (±0.007)	0.930 (±0.006)	0.995 (±0.001)

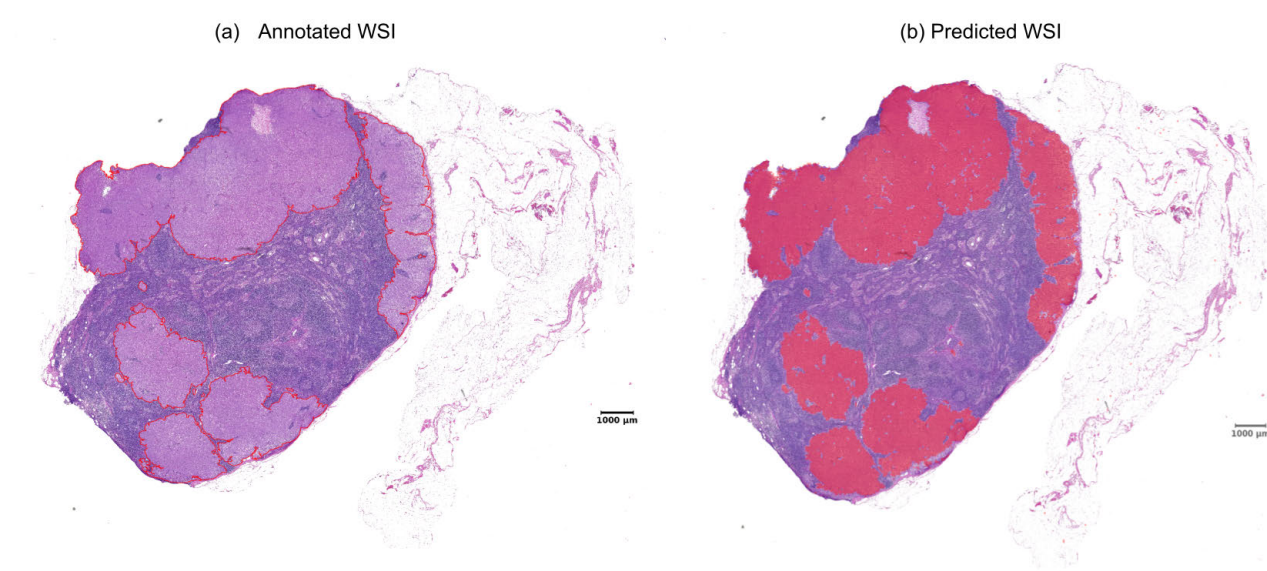


FIGURE 5. Qualitative analysis of breast cancer whole slide images (WSIs) using multilayer perceptron (MLP) adapters, fine-tuned with the iBOT model and the BDCSPN algorithm. The model was fine-tuned using 20 patches per class from the Camelyon17 Wilds dataset. The red area represents the predicted tumor mask.

computational resources or large-scale retraining should also be investigated.

We aim to expand the investigation to include a broader range of datasets, models, and methods and plan to explore the influence of data augmentation and preprocessing techniques on FSL in medical imaging. We aim to pursue the following directions to enhance clinical applicability:

- **The FSL method in personal therapeutics:** Personalized medicine aims to optimize treatment by identifying biomarkers that help tailor therapies to each patient [68].

The FSL method has demonstrated the ability to transfer drug or gene deletion response predictions across tissues, enabling DL models to predict cellular responses with limited training data [69]. This approach has the potential to integrate clinical metadata and genomic profiles with histopathology images by applying hybrid multimodal DL networks to enrich prototypical representations, make patient-specific diagnostic decisions, and achieve personalized treatments. For future work, we plan to investigate such hybrid DL models and multimodal data integrations in FSL.

- **Dynamic adapter architectures:** Recent studies on fine-tuning strategies have demonstrated the potential for rapid model adaptation across diverse tasks [18], [20], [21]. We aim to develop self-scaling adapters in future studies that dynamically adjust their complexity based on task difficulty, such as differentiating between rare and common tumor classes. In addition, we intend to design network architectures to generate attention maps that link class prototypes to critical morphological features, enhancing model interpretability and fostering clinician trust via spatial explanations.
- **Continuous learning:** Continuous model adaptation is essential to address rare or newly emerging diseases. We aim to investigate the use of residual connections to mitigate catastrophic forgetting during incremental adaptation, ensuring robust and stable model performance. Additionally, we plan to evaluate the proposed framework for various tumor classification tasks in clinical settings. A comprehensive evaluation across diverse medical classification scenarios is crucial before deploying the proposed method in real-world clinical practice.
- **Advanced model adaptation:** This study employs the BT, MoCo, and SwAV models based on ResNet50 and the iBOT model based on ViT-B, which have a relatively limited number of parameters (Section IV-B). More advanced and significantly larger foundation models explicitly designed for histopathology are emerging, some of which were released during this study [17], [70], [71]. However, due to time and resource constraints, these larger models could not be included in this study. For future work, we plan to investigate how large models could be adopted to improve the performance of histopathological image analysis.

VII. CONCLUSION

This study examines the efficacy of foundation models and adapter networks for efficient histopathology classification. The performance of the MLP and Conv adapters was promising when fine-tuned using diverse FSL algorithms to enable various foundation models to perform multiple histopathology classification tasks. The results highlight the influence of dataset characteristics, adapter types, and algorithms on model performance, particularly in data-constrained scenarios, as detailed in the results.

This study contributes to the ongoing trend of developing hybrid models that integrate foundational representations and task-specific adaptations. Specifically, the experiments revealed that the iBOT model demonstrated high classification accuracy when combined with MLP adapters across multiclass and binary classification tasks in medical imaging with

prototypical few-shot algorithms. The superior performance of the MLP adapters, especially in low-data regimes, suggests their robustness in handling the complexity of histopathology images. In addition, unlike conventional FSL studies focused on accuracy in few-shot settings, this study emphasizes the accuracy of histopathology classification, a more clinically relevant metric. This strategic shift was made to align with real-world healthcare applications, where reliability in decision-making and diagnostic accuracy are paramount. This technology has the potential for rapid adaptation to emerging diseases, personalized treatment planning, and deployment in resource-limited settings without the need for extensive retraining.

In conclusion, this study presents a novel adapter-based approach for adapting foundation models by using prototypical FSL algorithms in medical imaging. We provide an in-depth overview of histopathology classification with the potential to expand to more complex and diverse clinical tasks, including patient-specific tuning and deployment in real-world applications. We hope to contribute to the development of AI-driven precision oncology by addressing the current limitations and expanding on future research directions.

SUPPLEMENTARY MATERIALS

A. DETAILED CONVOLUTIONAL ADAPTER PERFORMANCE

See Fig. 6.

B. FINE-TUNING AND VALIDATION STAGE ACCURACY AND LOSS CURVES

See Fig. 7.

C. QUALITATIVE ANALYSIS OF BREAST CANCER WHOLE SLIDE IMAGES

See Fig. 8.

D. QUANTITATIVE DOWNSTREAM PERFORMANCE OF CAMELYON17 WILDS DATASET ON THE iBOT ENCODER

See Table 10.

E. QUANTITATIVE DOWNSTREAM PERFORMANCE OF NCT DATASET ON THE iBOT ENCODER

See Table 11.

ACKNOWLEDGMENT

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An earlier version of this paper was presented in part at the ASDP 2024 Congress in Seoul [72].

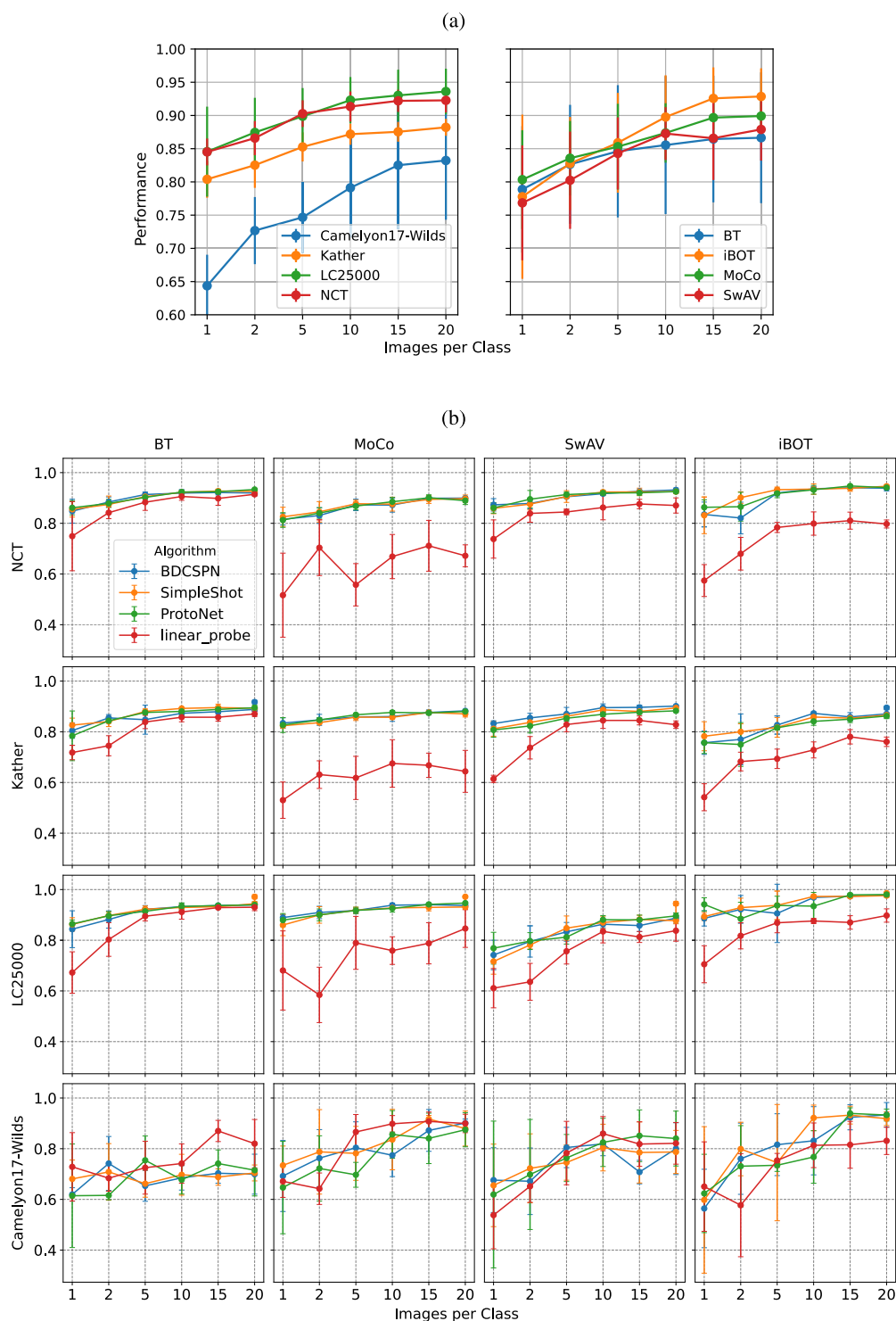


FIGURE 6. Histopathology classification performance of the convolutional (Conv) adapter. (a) Average classification accuracy of Conv adapters across datasets (left) and models (right). (b) Detailed average classification accuracy of Conv adapters when fine-tuned using various algorithms across all datasets.

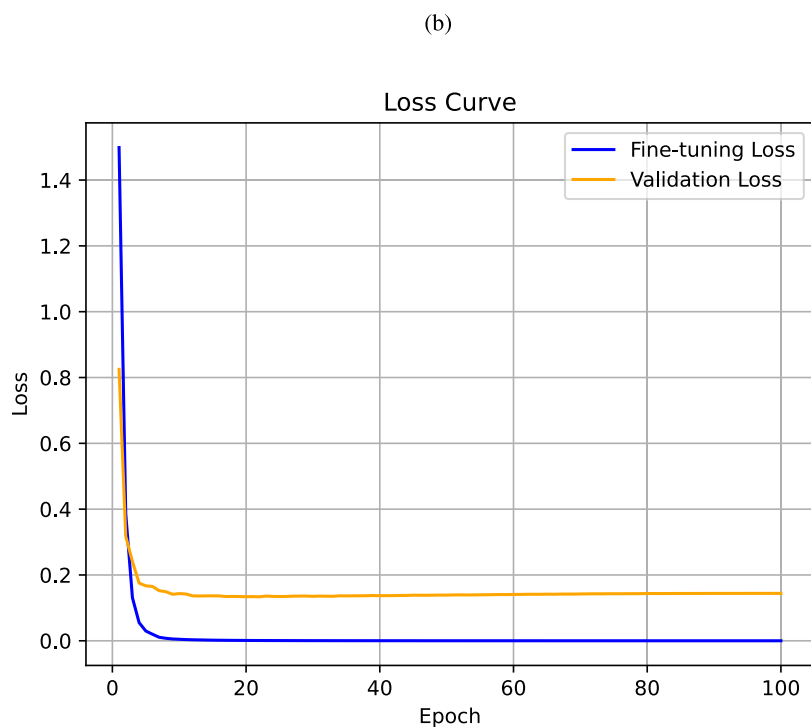
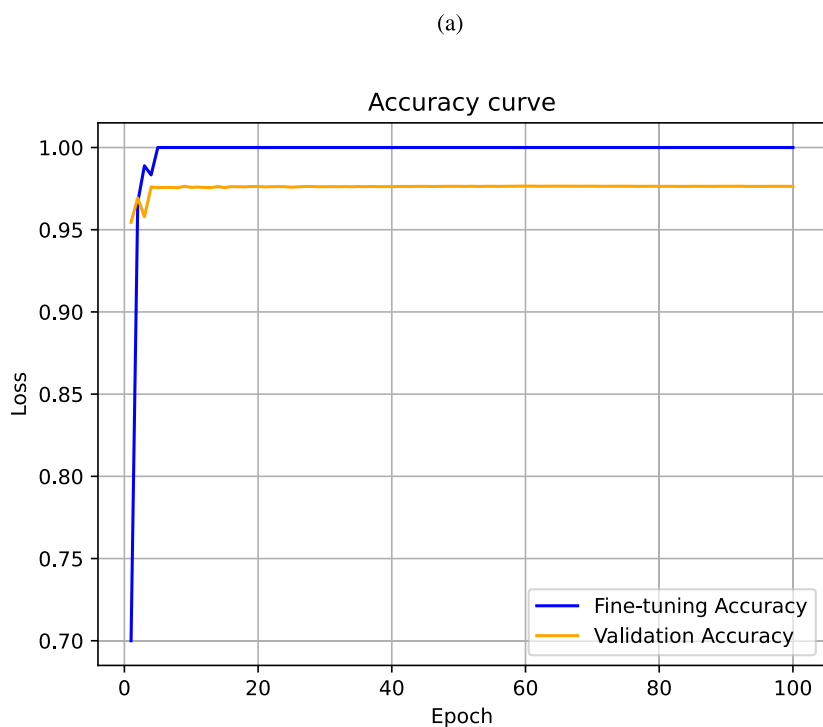


FIGURE 7. Fine-tuning and validation performance of the iBOT multilayer perceptron (MLP) adapter on the NCT dataset. (a) Accuracy and (b) loss curves are presented for the classifier fine-tuned using the SimpleShot algorithm with 20 patch images per class. Despite the limited number of images, the close alignment between the fine-tuning and validation curves in both panels indicates that the model generalizes well, with no significant overfitting.

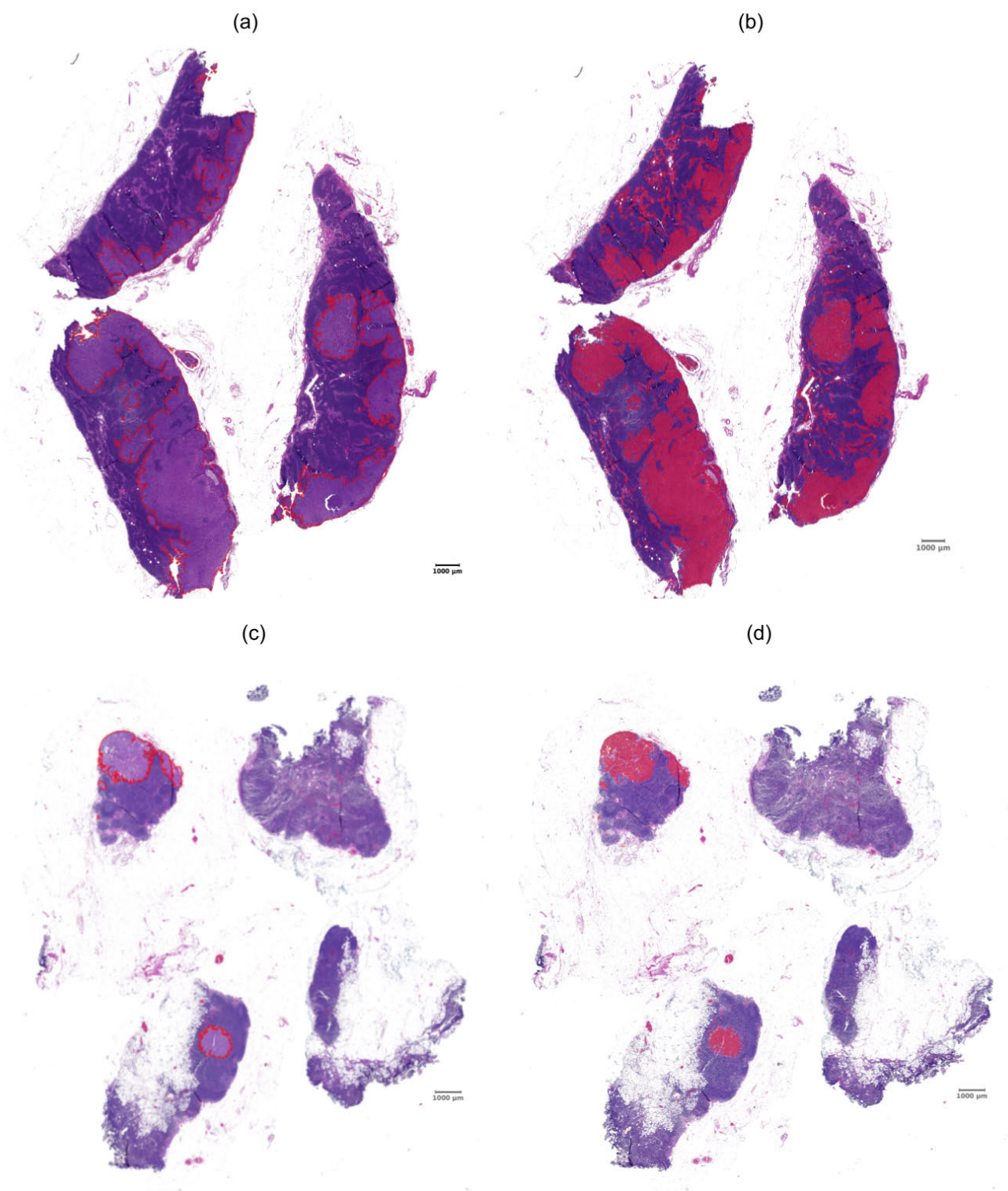


FIGURE 8. Qualitative analysis of breast cancer whole slide images (WSIs) using multilayer perceptron (MLP) adapters fine-tuned via the iBOT model and the BDCSPN algorithm. The model was fine-tuned using 20 patches per class from the Camelyon17 Wilds dataset. Panels (a) and (c) show annotated WSIs, while panels (b) and (d) illustrate model predictions, with tumor regions highlighted in red.

TABLE 10. Quantitative performance analysis of Camelyon17 Wilds using the iBOT model by numbers of images and adapters.

No. of images	Algorithm	Adapter	Accuracy	Precision	Recall	F1-score	AUC
1	BDCSPN	Conv MLP	0.541 ($\pm$ 0.14) 0.610 ($\pm$ 0.12)	0.561 ($\pm$ 0.183) 0.665 ($\pm$ 0.142)	0.541 ($\pm$ 0.14) 0.610 ($\pm$ 0.12)	0.475 ($\pm$ 0.167) 0.537 ($\pm$ 0.178)	0.595 ($\pm$ 0.209) 0.758 ($\pm$ 0.167)
	ProtoNet	Conv MLP	0.589 ($\pm$ 0.118) 0.758 ($\pm$ 0.179)	0.686 ($\pm$ 0.136) 0.751 ($\pm$ 0.247)	0.589 ($\pm$ 0.118) 0.758 ($\pm$ 0.179)	0.524 ($\pm$ 0.159) 0.731 ($\pm$ 0.231)	0.792 ($\pm$ 0.161) 0.836 ($\pm$ 0.236)
	SimpleShot	Conv MLP	0.604 ($\pm$ 0.096) 0.656 ($\pm$ 0.151)	0.716 ($\pm$ 0.077) 0.783 ($\pm$ 0.045)	0.604 ($\pm$ 0.096) 0.656 ($\pm$ 0.151)	0.538 ($\pm$ 0.160) 0.585 ($\pm$ 0.222)	0.843 ($\pm$ 0.130) 0.921 ($\pm$ 0.050)
2	BDCSPN	Conv MLP	0.735 ($\pm$ 0.141) 0.658 ($\pm$ 0.154)	0.803 ($\pm$ 0.083) 0.761 ($\pm$ 0.099)	0.735 ($\pm$ 0.141) 0.658 ($\pm$ 0.154)	0.705 ($\pm$ 0.174) 0.598 ($\pm$ 0.218)	0.905 ($\pm$ 0.068) 0.865 ($\pm$ 0.141)
	ProtoNet	Conv MLP	0.789 ($\pm$ 0.100) 0.782 ($\pm$ 0.064)	0.800 ($\pm$ 0.102) 0.815 ($\pm$ 0.031)	0.789 ($\pm$ 0.100) 0.782 ($\pm$ 0.064)	0.787 ($\pm$ 0.100) 0.774 ($\pm$ 0.077)	0.859 ($\pm$ 0.100) 0.922 ($\pm$ 0.035)
	SimpleShot	Conv MLP	0.750 ($\pm$ 0.173) 0.757 ($\pm$ 0.150)	0.751 ($\pm$ 0.181) 0.841 ($\pm$ 0.065)	0.750 ($\pm$ 0.173) 0.757 ($\pm$ 0.150)	0.728 ($\pm$ 0.213) 0.723 ($\pm$ 0.195)	0.830 ($\pm$ 0.153) 0.959 ($\pm$ 0.022)
5	BDCSPN	Conv MLP	0.899 ($\pm$ 0.056) 0.719 ($\pm$ 0.174)	0.907 ($\pm$ 0.052) 0.797 ($\pm$ 0.102)	0.899 ($\pm$ 0.056) 0.719 ($\pm$ 0.174)	0.899 ($\pm$ 0.056) 0.676 ($\pm$ 0.236)	0.962 ($\pm$ 0.031) 0.872 ($\pm$ 0.112)
	ProtoNet	Conv MLP	0.810 ($\pm$ 0.077) 0.910 ($\pm$ 0.041)	0.830 ($\pm$ 0.072) 0.921 ($\pm$ 0.030)	0.810 ($\pm$ 0.077) 0.910 ($\pm$ 0.041)	0.807 ($\pm$ 0.080) 0.909 ($\pm$ 0.042)	0.914 ($\pm$ 0.058) 0.978 ($\pm$ 0.014)
	SimpleShot	Conv MLP	0.742 ($\pm$ 0.221) 0.883 ($\pm$ 0.045)	0.792 ($\pm$ 0.192) 0.901 ($\pm$ 0.032)	0.742 ($\pm$ 0.221) 0.883 ($\pm$ 0.045)	0.675 ($\pm$ 0.311) 0.881 ($\pm$ 0.048)	0.802 ($\pm$ 0.240) 0.967 ($\pm$ 0.008)
10	BDCSPN	Conv MLP	0.903 ($\pm$ 0.072) 0.885 ($\pm$ 0.142)	0.916 ($\pm$ 0.050) 0.918 ($\pm$ 0.078)	0.903 ($\pm$ 0.072) 0.885 ($\pm$ 0.142)	0.901 ($\pm$ 0.075) 0.874 ($\pm$ 0.168)	0.966 ($\pm$ 0.022) 0.983 ($\pm$ 0.011)
	ProtoNet	Conv MLP	0.804 ($\pm$ 0.090) 0.917 ($\pm$ 0.056)	0.825 ($\pm$ 0.072) 0.923 ($\pm$ 0.050)	0.804 ($\pm$ 0.090) 0.917 ($\pm$ 0.056)	0.799 ($\pm$ 0.096) 0.916 ($\pm$ 0.057)	0.922 ($\pm$ 0.041) 0.979 ($\pm$ 0.016)
	SimpleShot	Conv MLP	0.908 ($\pm$ 0.081) 0.933 ($\pm$ 0.042)	0.927 ($\pm$ 0.051) 0.938 ($\pm$ 0.033)	0.908 ($\pm$ 0.081) 0.933 ($\pm$ 0.042)	0.905 ($\pm$ 0.087) 0.933 ($\pm$ 0.043)	0.984 ($\pm$ 0.008) 0.980 ($\pm$ 0.008)
15	BDCSPN	Conv MLP	0.950 ($\pm$ 0.006) 0.920 ($\pm$ 0.013)	0.953 ($\pm$ 0.005) 0.929 ($\pm$ 0.008)	0.950 ($\pm$ 0.006) 0.920 ($\pm$ 0.013)	0.950 ($\pm$ 0.006) 0.920 ($\pm$ 0.014)	0.983 ($\pm$ 0.008) 0.976 ($\pm$ 0.009)
	ProtoNet	Conv MLP	0.938 ($\pm$ 0.014) 0.934 ($\pm$ 0.026)	0.942 ($\pm$ 0.013) 0.941 ($\pm$ 0.018)	0.938 ($\pm$ 0.014) 0.934 ($\pm$ 0.026)	0.937 ($\pm$ 0.014) 0.934 ($\pm$ 0.026)	0.980 ($\pm$ 0.011) 0.981 ($\pm$ 0.010)
	SimpleShot	Conv MLP	0.945 ($\pm$ 0.015) 0.839 ($\pm$ 0.193)	0.948 ($\pm$ 0.014) 0.889 ($\pm$ 0.104)	0.945 ($\pm$ 0.015) 0.839 ($\pm$ 0.193)	0.945 ($\pm$ 0.016) 0.805 ($\pm$ 0.266)	0.981 ($\pm$ 0.011) 0.944 ($\pm$ 0.064)
20	BDCSPN	Conv MLP	0.925 ($\pm$ 0.029) 0.939 ($\pm$ 0.017)	0.935 ($\pm$ 0.021) 0.944 ($\pm$ 0.016)	0.925 ($\pm$ 0.029) 0.939 ($\pm$ 0.017)	0.924 ($\pm$ 0.029) 0.939 ($\pm$ 0.017)	0.979 ($\pm$ 0.012) 0.980 ($\pm$ 0.015)
	ProtoNet	Conv MLP	0.938 ($\pm$ 0.012) 0.958 ($\pm$ 0.010)	0.944 ($\pm$ 0.009) 0.959 ($\pm$ 0.010)	0.938 ($\pm$ 0.012) 0.958 ($\pm$ 0.010)	0.938 ($\pm$ 0.012) 0.958 ($\pm$ 0.010)	0.980 ($\pm$ 0.007) 0.986 ($\pm$ 0.008)
	SimpleShot	Conv MLP	0.923 ($\pm$ 0.025) 0.877 ($\pm$ 0.142)	0.933 ($\pm$ 0.018) 0.908 ($\pm$ 0.085)	0.923 ($\pm$ 0.025) 0.877 ($\pm$ 0.142)	0.923 ($\pm$ 0.026) 0.866 ($\pm$ 0.166)	0.981 ($\pm$ 0.006) 0.955 ($\pm$ 0.069)

TABLE 11. Quantitative performance analysis of NCT using the iBOT model by numbers of images and adapters.

No. of images	Algorithm	Adapter	Accuracy	Precision	Recall	F1-score	AUC
1	BDCSPN	Conv MLP	0.556 ($\pm$ 0.228) 0.787 ($\pm$ 0.242)	0.685 ($\pm$ 0.161) 0.867 ($\pm$ 0.026)	0.528 ($\pm$ 0.210) 0.736 ($\pm$ 0.264)	0.471 ($\pm$ 0.255) 0.725 ($\pm$ 0.270)	0.911 ($\pm$ 0.089) 0.959 ($\pm$ 0.046)
	ProtoNet	Conv MLP	0.733 ($\pm$ 0.062) 0.714 ($\pm$ 0.173)	0.765 ($\pm$ 0.032) 0.756 ($\pm$ 0.083)	0.702 ($\pm$ 0.065) 0.663 ($\pm$ 0.168)	0.686 ($\pm$ 0.068) 0.642 ($\pm$ 0.175)	0.950 ($\pm$ 0.026) 0.946 ($\pm$ 0.039)
	SimpleShot	Conv MLP	0.624 ($\pm$ 0.336) 0.779 ($\pm$ 0.089)	0.624 ($\pm$ 0.341) 0.768 ($\pm$ 0.074)	0.587 ($\pm$ 0.312) 0.750 ($\pm$ 0.096)	0.572 ($\pm$ 0.311) 0.719 ($\pm$ 0.093)	0.839 ($\pm$ 0.236) 0.958 ($\pm$ 0.017)
2	BDCSPN	Conv MLP	0.634 ($\pm$ 0.252) 0.817 ($\pm$ 0.120)	0.710 ($\pm$ 0.146) 0.835 ($\pm$ 0.053)	0.579 ($\pm$ 0.253) 0.783 ($\pm$ 0.113)	0.546 ($\pm$ 0.282) 0.773 ($\pm$ 0.127)	0.879 ($\pm$ 0.186) 0.973 ($\pm$ 0.017)
	ProtoNet	Conv MLP	0.712 ($\pm$ 0.347) 0.897 ($\pm$ 0.037)	0.754 ($\pm$ 0.157) 0.851 ($\pm$ 0.052)	0.684 ($\pm$ 0.316) 0.867 ($\pm$ 0.038)	0.660 ($\pm$ 0.341) 0.852 ($\pm$ 0.050)	0.901 ($\pm$ 0.167) 0.986 ($\pm$ 0.007)
	SimpleShot	Conv MLP	0.855 ($\pm$ 0.048) 0.863 ($\pm$ 0.075)	0.849 ($\pm$ 0.029) 0.840 ($\pm$ 0.074)	0.820 ($\pm$ 0.057) 0.823 ($\pm$ 0.079)	0.806 ($\pm$ 0.059) 0.807 ($\pm$ 0.094)	0.977 ($\pm$ 0.009) 0.977 ($\pm$ 0.015)
5	BDCSPN	Conv MLP	0.927 ($\pm$ 0.015) 0.842 ($\pm$ 0.219)	0.898 ($\pm$ 0.016) 0.875 ($\pm$ 0.093)	0.903 ($\pm$ 0.017) 0.821 ($\pm$ 0.224)	0.898 ($\pm$ 0.017) 0.805 ($\pm$ 0.250)	0.987 ($\pm$ 0.003) 0.976 ($\pm$ 0.039)
	ProtoNet	Conv MLP	0.861 ($\pm$ 0.106) 0.939 ($\pm$ 0.015)	0.865 ($\pm$ 0.046) 0.915 ($\pm$ 0.018)	0.828 ($\pm$ 0.100) 0.912 ($\pm$ 0.028)	0.816 ($\pm$ 0.120) 0.911 ($\pm$ 0.023)	0.984 ($\pm$ 0.007) 0.992 ($\pm$ 0.003)
	SimpleShot	Conv MLP	0.929 ($\pm$ 0.025) 0.939 ($\pm$ 0.003)	0.910 ($\pm$ 0.021) 0.909 ($\pm$ 0.004)	0.912 ($\pm$ 0.019) 0.914 ($\pm$ 0.005)	0.907 ($\pm$ 0.022) 0.910 ($\pm$ 0.004)	0.992 ($\pm$ 0.003) 0.993 ($\pm$ 0.001)
10	BDCSPN	Conv MLP	0.750 ($\pm$ 0.357) 0.952 ($\pm$ 0.007)	0.806 ($\pm$ 0.224) 0.929 ($\pm$ 0.012)	0.731 ($\pm$ 0.342) 0.932 ($\pm$ 0.011)	0.718 ($\pm$ 0.357) 0.928 ($\pm$ 0.012)	0.949 ($\pm$ 0.092) 0.995 ($\pm$ 0.001)
	ProtoNet	Conv MLP	0.919 ($\pm$ 0.072) 0.943 ($\pm$ 0.008)	0.910 ($\pm$ 0.042) 0.913 ($\pm$ 0.014)	0.894 ($\pm$ 0.080) 0.918 ($\pm$ 0.012)	0.894 ($\pm$ 0.078) 0.914 ($\pm$ 0.014)	0.989 ($\pm$ 0.010) 0.994 ($\pm$ 0.002)
	SimpleShot	Conv MLP	0.867 ($\pm$ 0.178) 0.938 ($\pm$ 0.007)	0.891 ($\pm$ 0.070) 0.907 ($\pm$ 0.010)	0.832 ($\pm$ 0.206) 0.910 ($\pm$ 0.009)	0.820 ($\pm$ 0.228) 0.906 ($\pm$ 0.011)	0.987 ($\pm$ 0.015) 0.994 ($\pm$ 0.002)
15	BDCSPN	Conv MLP	0.946 ($\pm$ 0.003) 0.953 ($\pm$ 0.006)	0.922 ($\pm$ 0.003) 0.929 ($\pm$ 0.008)	0.927 ($\pm$ 0.005) 0.934 ($\pm$ 0.011)	0.923 ($\pm$ 0.003) 0.931 ($\pm$ 0.009)	0.994 ($\pm$ 0.002) 0.995 ($\pm$ 0.003)
	ProtoNet	Conv MLP	0.943 ($\pm$ 0.007) 0.953 ($\pm$ 0.003)	0.917 ($\pm$ 0.005) 0.928 ($\pm$ 0.003)	0.922 ($\pm$ 0.008) 0.932 ($\pm$ 0.005)	0.918 ($\pm$ 0.006) 0.929 ($\pm$ 0.004)	0.994 ($\pm$ 0.001) 0.996 ($\pm$ 0.001)
	SimpleShot	Conv MLP	0.944 ($\pm$ 0.003) 0.952 ($\pm$ 0.004)	0.919 ($\pm$ 0.004) 0.927 ($\pm$ 0.006)	0.923 ($\pm$ 0.004) 0.930 ($\pm$ 0.005)	0.919 ($\pm$ 0.004) 0.927 ($\pm$ 0.006)	0.993 ($\pm$ 0.001) 0.995 ($\pm$ 0.001)
20	BDCSPN	Conv MLP	0.947 ($\pm$ 0.005) 0.950 ($\pm$ 0.012)	0.921 ($\pm$ 0.008) 0.927 ($\pm$ 0.016)	0.927 ($\pm$ 0.009) 0.929 ($\pm$ 0.014)	0.923 ($\pm$ 0.008) 0.926 ($\pm$ 0.016)	0.994 ($\pm$ 0.002) 0.996 ($\pm$ 0.001)
	ProtoNet	Conv MLP	0.944 ($\pm$ 0.006) 0.954 ($\pm$ 0.004)	0.919 ($\pm$ 0.005) 0.929 ($\pm$ 0.006)	0.923 ($\pm$ 0.005) 0.934 ($\pm$ 0.008)	0.920 ($\pm$ 0.005) 0.931 ($\pm$ 0.007)	0.995 ($\pm$ 0.001) 0.995 ($\pm$ 0.001)
	SimpleShot	Conv MLP	0.950 ($\pm$ 0.003) 0.955 ($\pm$ 0.002)	0.923 ($\pm$ 0.007) 0.928 ($\pm$ 0.005)	0.930 ($\pm$ 0.005) 0.935 ($\pm$ 0.003)	0.926 ($\pm$ 0.006) 0.931 ($\pm$ 0.004)	0.995 ($\pm$ 0.001) 0.996 ($\pm$ 0.001)

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